A Constructive Introduction to Quantum Field Theory

Lecture 4: Forward–backward equations

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Abstract

Building on the scale decomposition and the stochastic observables of Lecture 3, we introduce interactions. The interacting measure can be generated in three ways: as a renormalisation group decomposition of the covariance, as an optimal control problem, whose dynamic programming equation is the Polchinski equation, and as a filtering problem, where the regulator is the information gained on the terminal field. The three views meet in one forward–backward system for a field and a force, and its natural coordinates are the force coordinates: the forward equation loses its drift, and the conditional covariance becomes the propagator. All this can be said in terms of forces, propagators and covariances. In the gradient case there is a parallel story for potentials: the value function generates the canonical transformation to force coordinates, the Legendre transform of the log-partition function of the filter gives the effective average action, and its flow is the Wetterich equation. We then discuss stochastic observables, the field equation at every resolution, whose noise is a filtered Gaussian noise and therefore sub-Gaussian, and the passage to infinite volume. The appendices contain the algebra of the flow, perturbation theory and counterterms, and the functional inequalities that follow from the localisation. The treatment is formal: we compute as if all objects were smooth and integrable, and we indicate where the real difficulties lie.

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Introduction

In Lecture 3 we decomposed the Gaussian free field along a resolution parameter and took as primary objects the stochastic observables: martingales in the filtration of finite-resolution measurements [BCG26]. In this lecture we introduce interactions, and we tell one story from three points of view: renormalisation, optimal control and filtering. See also [Pol84, BK87, Sal99, BBD24] for the renormalisation group side and [BG20, BG23] for the variational side.

Structure. Section 7 generates the interacting measure in the three ways, and shows that they produce the same field and the same force. Section 8 finds the natural coordinates, first in terms of forces, propagators and covariances, then in terms of potentials, where the value function generates the canonical transformation and the Legendre transform leads to the Wetterich equation. Section 9 treats observables and the field equation with its sub-Gaussian noise. Appendices A and B are additional material, referred to from the main text where needed.

What is formal. Rigorous statements require a finite-dimensional setting: a finite lattice, or a Fourier cutoff on the torus, together with a finite terminal resolution T. There, all the identities below are theorems under mild growth assumptions on the interaction, and their proofs are the computations we give. The hard problems are the limits $T \to \infty$ (ultraviolet) and infinite volume (infrared), together with the locality of the limiting objects. We mark with (formal) the statements that we only derive at this level.

Conventions. As in Lectures 1–3: $\mathcal{M} = \mathbb{T}_L^d$ or $\mathbb{R}^d$, $A = m^2 - \Delta$, $C = A^{-1}$ with kernel G, μ_m the GFF, $\langle f, g\rangle = \int fg$ for real functions. For a functional F on smooth fields, $\mathrm{D}F(\varphi) \in \mathcal{S}'(\mathcal{M})$ and $\mathrm{D}^2F(\varphi)$ are its first and second derivatives, $D_h F = \langle \mathrm{D}F, h\rangle$. $H_n(a;c)$ are the Hermite polynomials of variance c, and μ^Q is the centred Gaussian measure with covariance Q.

Main lecture

7 Three ways to generate the interacting measure

Fix a finite volume $\mathcal{M} = \mathbb{T}_L^d$ and a terminal resolution $T < \infty$, which plays the role of an ultraviolet cutoff: the reference field $X_T \sim \mu^{Q_T}$ is smooth. Let V_T be an interaction, a functional of smooth fields bounded below. Our main example is

$$V_T(\varphi) = \int_{\mathbb{T}_L^d} [\lambda H_4(\varphi(x); c_T) + \gamma_T H_2(\varphi(x); c_T)]\mathrm{d}x, \tag{7.1}$$

with a mass counterterm γ_T to be fixed in Appendix A.2 ($\gamma_T = 0$ for $d = 2$). The interacting measure at cutoff T is

$$\nu_T(\mathrm{d}\varphi) = Z_T^{-1} \mathrm{e}^{-V_T(\varphi)} \mu^{Q_T}(\mathrm{d}\varphi). \tag{7.2}$$

How does the Gaussian scale decomposition of Lecture 3 generate ν_T? There are three answers.

- **Renormalisation** (Section 7.1). Split the covariance into shells and integrate them out one at a time. At every resolution the field is again a Gibbs measure, and it is driven by an effective force.
- **Optimal control** (Section 7.2). Steer the Gaussian field by a drift, at the least cost. The effective potential is the value function, and the Polchinski equation is the dynamic programming equation.
- **Filtering** (Section 7.3). Measure the terminal field with increasing precision. The law of the terminal field given the measurement is the posterior, and the regulator is the information gained.

Section 7.4 shows that the three constructions produce the same process. Throughout,

$$S_t := Q_T - Q_t = Q_{T,t}, \qquad \dot{S}_t = -\dot{Q}_t, \qquad S_T = 0,$$

is the covariance still unresolved at resolution t.

7.1 Renormalisation: decomposing the covariance

Decompose the covariance as in Lecture 3: $Q_T = Q_t + S_t$, and correspondingly

$$X_T = X_t + (X_T - X_t)$$

with independent terms. Integrating out the second one defines the effective potential.

Definition 7.1. *The* effective potential *at resolution $t \in [0,T]$ is*

$$\mathrm{e}^{-V_t(\varphi)} := \int \mathrm{e}^{-V_T(\varphi+\zeta)}\, \mu^{Q_{T,t}}(\mathrm{d}\zeta). \tag{7.3}$$

V_t is the interaction felt by the field at resolution t once the fluctuations between t and T have been integrated out: it is the Wilson effective action. At $t=0$, $V_0(0) = -\log Z_T$ is the free energy. Here is the first key fact.

Proposition 7.2. (Marginals) *Let $\mathbb{P}$ be defined by*

$$\frac{\mathrm{d}\mathbb{P}}{\mathrm{d}\mathbb{P}_0} = Z_T^{-1}\mathrm{e}^{-V_T(X_T)},$$

where $\mathbb{P}_0$ is the law of the Gaussian process X. Then $\mathrm{Law}_{\mathbb{P}}(X_T) = \nu_T$, and for every $t \leqslant T$

$$\mathrm{Law}_{\mathbb{P}}(X_t) = \nu_t, \qquad \nu_t(\mathrm{d}\varphi) := Z_T^{-1}\mathrm{e}^{-V_t(\varphi)}\mu^{Q_t}(\mathrm{d}\varphi). \tag{7.4}$$

Moreover, the conditional law of X_T given $\mathcal{F}_t$ is

$$\mu_t^{X_t}(\mathrm{d}\varphi), \qquad \mu_t^{\phi}(G) := \mathrm{e}^{V_t(\phi)} \int G(\phi+\zeta)\, \mathrm{e}^{-V_T(\phi+\zeta)}\, \mu^{Q_{T,t}}(\mathrm{d}\zeta). \tag{7.5}$$

Proof. Under $\mathbb{P}_0$, $X_T = X_t + (X_T - X_t)$ with an independent increment of law $\mu^{Q_{T,t}}$. Hence $\mathbb{E}_{\mathbb{P}}[G(X_t)] = Z_T^{-1}\,\mathbb{E}_0[G(X_t)\mathrm{e}^{-V_T(X_T)}] = Z_T^{-1}\,\mathbb{E}_0[G(X_t)\mathrm{e}^{-V_t(X_t)}]$. The conditional law follows in the same way. □

So at every resolution the field is again a Gibbs measure, with reference covariance Q_t and potential V_t. The measure-valued martingale $(\mu_t^{X_t})_t$ goes from ν_T at $t=0$ to δ_{X_T} at $t=T$: it is a *stochastic localisation* of ν_T in the sense of Eldan [Eld23], adapted to the covariance profile [BBD24, BCG26].

The field as a process. The density of the interacting law with respect to the Gaussian one is a martingale in resolution, so by Girsanov the field acquires a drift, and this drift is the effective force. The resulting process is the **Polchinski process**.

Theorem 7.3. (FBSDE for the interacting field) (formal) *Under $\mathbb{P}$ the process $\Phi_t := X_t$, the* interacting field at resolution t, *solves*

$$\mathrm{d}\Phi_t = \dot{Q}_t\, F_t\, \mathrm{d}t + \dot{Q}_t^{1/2}\, \mathrm{d}W_t^{\mathbb{P}}, \qquad F_t = -\mathrm{D}V_t(\Phi_t) = \mathbb{E}_t^{\mathbb{P}}\,[-\mathrm{D}V_T(\Phi_T)], \tag{7.6}$$

with $\Phi_0 = 0$ and a $\mathbb{P}$-cylindrical Brownian motion $W^{\mathbb{P}}$. In particular the effective force *$(F_t)_{t\leqslant T}$ is a $\mathbb{P}$-martingale: an observable field of the interacting theory.*

Proof. The density process is $M_t := \mathbb{E}_0[\,\mathrm{d}\mathbb{P}\,/\,\mathrm{d}\mathbb{P}_0|\mathcal{F}_t] = Z_T^{-1}\mathrm{e}^{-V_t(X_t)}$, by Proposition 7.2. It is a martingale, so the drift in Itô's formula vanishes and $\mathrm{d}M_t = -M_t\langle \mathrm{D}V_t(X_t), \dot{Q}_t^{1/2}\,\mathrm{d}W_t\rangle$. By Girsanov's theorem, $W_t^{\mathbb{P}} = W_t + \int_0^t \dot{Q}_s^{1/2}\mathrm{D}V_s(X_s)\,\mathrm{d}s$ is a $\mathbb{P}$-Brownian motion, which is (7.6). Differentiating (7.3) gives $\mathrm{D}V_t(\phi) = \mu_t^{\phi}(\mathrm{D}V_T)$, which is the second identity. □

Read as an equation for an unknown pair (Φ, F), (7.6) is a *forward–backward* system [MY99]. The field is driven forward in resolution by a force that is determined backward, as the conditional expectation of a functional of the terminal field:

$$\Phi_t = \int_0^t \dot{Q}_s F_s\,\mathrm{d}s + \int_0^t \dot{Q}_s^{1/2}\mathrm{d}W_s, \qquad F_t = \mathbb{E}_t\,[-\mathrm{D}V_T(\Phi_T)], \qquad t \in [0,T]. \tag{7.7}$$

Here and below we drop the superscript and write W for the Brownian motion driving the interacting field. No equation for the potential is needed to state (7.7), and $\nu_T = \mathrm{Law}(\Phi_T)$ is recovered from the solution. This is the starting point of FBSDE constructions of Euclidean fields (Figure 7.1): one solves (7.7) with renormalised V_T, with estimates uniform in T and in the volume, and takes the limit. The field coordinate Φ is *not* an observable, only a semi-martingale. Its martingale part is the Gaussian noise, and the observables are built from it and from F.

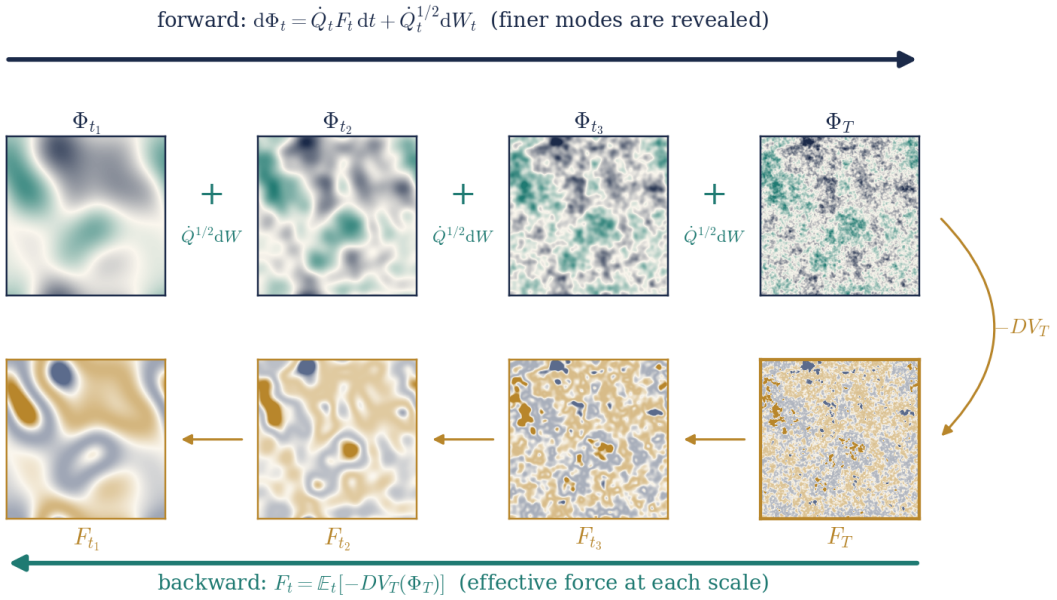


Figure 7.1. The forward–backward structure of (7.7) on $\mathbb{T}^2$, heat profile, at resolutions $t_1 < t_2 < t_3 < T$, all built from the same noise. Top: the field gains the fluctuation of each shell, $\dot{Q}_t^{1/2}\mathrm{d}W_t$, and reveals finer modes. Bottom: the effective force $F_t = \mathbb{E}_t[-\mathrm{D}V_T(\Phi_T)]$ for the Φ^4 interaction, determined backward from the terminal interaction. It is shown to first order in λ, where it is the Wick cube $-4\lambda H_3(\Phi_t; c_t)$, and each panel is in its own units.

7.2 Optimal control: the Polchinski equation

The measure ν_T also solves a minimisation problem. Perturb the Gaussian field by an adapted drift, and pay for the drift and for the interaction at the endpoint. The effective potential is the value function of this problem. Its equation in t, which we first derive directly from (7.3), is the dynamic programming equation.

Theorem 7.4. **(Polchinski equation [Pol84])** (formal) *V_t solves the backward Hamilton–Jacobi equation*

$$\partial_t V_t = \tfrac{1}{2}\langle \mathrm{D}V_t, \dot{Q}_t\,\mathrm{D}V_t\rangle - \tfrac{1}{2}\mathrm{Tr}(\dot{Q}_t\,\mathrm{D}^2 V_t), \qquad t \in [0,T], \tag{7.8}$$

with terminal condition V_T.

Proof. $u_t := \mathrm{e}^{-V_t} = \mathrm{e}^{\frac{1}{2}\Delta_{Q_{T,t}}} u_T$ is a Gaussian convolution, where $\Delta_Q := \mathrm{Tr}(Q\,\mathrm{D}^2)$. Since $\partial_t Q_{T,t} = -\dot{Q}_t$, it solves the backward heat equation $\partial_t u_t = -\frac{1}{2}\mathrm{Tr}(\dot{Q}_t\,\mathrm{D}^2 u_t)$. Insert $\mathrm{D}^2 u = (\mathrm{D}V \otimes \mathrm{D}V - \mathrm{D}^2 V)\,u$ and divide by $-u$. □

The nonlinear term generates new, generally nonlocal, interactions (trees), and the trace term contracts pairs of fields with the covariance of the shell $[t, t+\mathrm{d}t]$ (loops). The substitution $u = \mathrm{e}^{-V}$, which linearises the equation, is a Cole–Hopf transform: we come back to this in Section 8.1, and in Appendix A.

The control problem. Let $\mathbb{H}_a$ be the set of progressively measurable drifts $u\colon[0,T]\to L^2(M)$ with $\mathbb{E}\int_0^T\|u_s\|^2\mathrm{d}s<\infty$, and set

$$I_t(u):=\int_0^t \dot{Q}_s^{1/2}u_s\,\mathrm{d}s,\qquad X^u:=X+I(u),$$

the field driven by the control u under the Gaussian reference $\mathbb{P}_0$.

The following result is standard in the stochastic optimal control setting, see e.g. Fleming and Soner's book [FS06]. A useful non-Markovian generalisation is given by Boué–Dupuis [BD98].

Theorem 7.5. (Polchinski is HJB) (formal)

$$-\log\mathbb{E}_0[\mathrm{e}^{-V_T(X_T)}]=V_0(0)=\inf_{u\in\mathbb{H}_a}\mathbb{E}_0\Big[V_T(X_T^u)+\frac12\int_0^T\|u_s\|^2\mathrm{d}s\Big]. \tag{7.9}$$

The infimum is attained at the feedback control $u_s^*=-\dot{Q}_s^{1/2}\mathrm{D}V_s(X_s^{u^*})$, *and* $\mathrm{Law}(X_T^{u^*})=\nu_T$. *The effective potential is the value function, and* (7.8) *is the Hamilton–Jacobi–Bellman equation of this control problem.*

Proof. (Verification.) For any u, by Itô's formula and (7.8),

$$\begin{aligned}\mathrm{d}V_t(X_t^u)&=\Big[\tfrac12\|\dot{Q}_t^{1/2}\mathrm{D}V_t\|^2+\langle\dot{Q}_t^{1/2}\mathrm{D}V_t,u_t\rangle\Big]\mathrm{d}t+\mathrm{d}M_t\\&=\Big[\tfrac12\|u_t+\dot{Q}_t^{1/2}\mathrm{D}V_t\|^2-\tfrac12\|u_t\|^2\Big]\mathrm{d}t+\mathrm{d}M_t,\end{aligned}$$

with $\mathrm{D}V_t=\mathrm{D}V_t(X_t^u)$ and a martingale M. Integrating and taking expectations,

$$\mathbb{E}_0\Big[V_T(X_T^u)+\frac12\int_0^T\|u_t\|^2\mathrm{d}t\Big]=V_0(0)+\frac12\,\mathbb{E}_0\int_0^T\|u_t+\dot{Q}_t^{1/2}\mathrm{D}V_t(X_t^u)\|^2\mathrm{d}t. \tag{7.10}$$

So the cost is $\geqslant V_0(0)$, with equality iff $u=u^*$. With this control X^{u^*} solves the same SDE as Φ in (7.6). □

Euler–Lagrange equation = FBSDE. Perturbing $u\to u+\varepsilon v$ in (7.9), the first-order condition is $\mathbb{E}\int_0^T\langle u_s+\dot{Q}_s^{1/2}\,\mathbb{E}_s[\mathrm{D}V_T(X_T^u)],v_s\rangle\mathrm{d}s=0$ for all adapted v, i.e.

$$u_s=-\dot{Q}_s^{1/2}\,\mathbb{E}_s[\mathrm{D}V_T(X_T^u)]=\dot{Q}_s^{1/2}F_s. \tag{7.11}$$

This is (7.7). The FBSDE is the Euler–Lagrange equation of the variational problem, and the Polchinski equation is its Hamilton–Jacobi equation.

Entropy. By Girsanov, $\frac12\,\mathbb{E}_0\int\|u\|^2\geqslant\mathrm{Ent}(\mathrm{Law}(X^u)\,|\,\mathrm{Law}(X))$, which is at least the entropy of the terminal marginal. So (7.9) is a pathwise lift of the Gibbs variational principle $-\log Z_T=\inf_\nu\{\int V_T\,\mathrm{d}\nu+\mathrm{Ent}(\nu|\mu^{Q_T})\}$. The lift is useful because adapted drifts can be *constructed scale by scale*: one looks for u as a functional of the Gaussian process X, not as an unknown measure.

Generating functionals. Adding a test functional f,

$$\mathcal{W}_T(f):=-\log\int\mathrm{e}^{-f}\mathrm{d}\nu_T=\inf_u\mathbb{E}_0\Big[f(X_T^u)+V_T(X_T^u)+\frac12\int_0^T\|u_s\|^2\mathrm{d}s\Big]-V_0(0). \tag{7.12}$$

Bounds on $\mathcal{W}_T(f)$ uniform in T, for f in a rich class, give tightness, moment bounds and the identification of limits. For Φ^4_3 the naive problem does not have uniform bounds: the leading-order optimal drift $u_s\approx-4\lambda\,\dot{Q}_s^{1/2}{:}X_s^3{:}$ has a divergent cost. One renormalises by the change of variables $u=-4\lambda\,\dot{Q}^{1/2}{:}X^3{:}+r$. After it, and after further paracontrolled changes of variables, the singular terms are explicit polynomials in the Wick powers of X, cancelled by *the counterterms*, and the residual functional is coercive. Its control in Besov and paracontrolled norms, uniformly in T, is the construction of Φ^4_3 in [BG20]; see also [BG21].

Pontryagin's maximum principle. Consider first a deterministic problem: minimise the cost $J(u) = g(x_T) + \int_t^T \ell(x_s, u_s)\,\mathrm{d}s$ over controls u on $[t,T]$, where the state solves $\dot{x}_s = b(x_s, u_s)$, $x_t = x$. With the *Hamiltonian*

$$H(x,p,u) := \langle p, b(x,u)\rangle - \ell(x,u),$$

Pontryagin's principle [PBGM62] says that along an optimal pair (x,u) there is a *costate* p such that

$$\dot{x}_s = \partial_p H, \qquad \dot{p}_s = -\partial_x H, \qquad p_T = -\mathrm{D}g(x_T), \qquad u_s \in \arg\max_v H(x_s, p_s, v).$$

The state runs forward from its initial value, the costate backward from its terminal value: a two-point boundary value problem. In the smooth case the principle is the first-order condition. If $\xi = \partial_\varepsilon x$ is the variation of the state under $u \to u + \varepsilon v$, then $\dot{\xi} = \partial_x b\,\xi + \partial_u b\, v$, $\xi_t = 0$, and the costate equation is exactly what makes $\frac{\mathrm{d}}{\mathrm{d}s}\langle p, \xi\rangle = \langle \partial_x \ell, \xi\rangle + \langle p, \partial_u b\, v\rangle$. Integrating over $[t, T]$ and using the terminal condition,

$$\frac{\mathrm{d}}{\mathrm{d}\varepsilon} J(u + \varepsilon v)|_{\varepsilon=0} = -\int_t^T \langle \partial_u H(x_s, p_s, u_s), v_s\rangle\,\mathrm{d}s.$$

So the costate is the Lagrange multiplier of the constraint $\dot{x} = b$, and an optimal control is a critical point of the Hamiltonian.

The costate is a gradient. Let $V(t,x) = \min_u J(u)$ be the value function. Comparing the cost from (t,x) with the cost from $(t + \mathrm{d}t, x + b\,\mathrm{d}t)$ gives the Hamilton–Jacobi–Bellman equation

$$\partial_t V = H_*(x, -\mathrm{D}V), \qquad H_*(x,p) := \max_u H(x,p,u), \qquad V(T,\cdot) = g.$$

Its characteristics are Hamilton's equations for H_*, and along them $p_s = -\mathrm{D}V(s, x_s)$: the costate is the sensitivity of the optimal cost to a displacement of the state, i.e. the force that the future exerts on the present.

The Hamiltonian of the field. In the problem of Theorem 7.5 the state is the field, $b(x,u) = \dot{Q}_s^{1/2} u$, $\ell = \frac{1}{2}\|u\|^2$ and $g = V_T$. Then $H(p,u) = \langle p, \dot{Q}_s^{1/2} u\rangle - \frac{1}{2}\|u\|^2$ is maximal at $u = \dot{Q}_s^{1/2} p$, and

$$H_*(p) = \frac{1}{2}\langle p, \dot{Q}_s p\rangle$$

is the Hamiltonian of a free particle with mass matrix $\dot{Q}_s^{-1}$: it does not depend on the state. Hence $\dot{p} = 0$ and $\dot{x} = \dot{Q}_s p$: the costate is conserved, and from t to T the state moves along the linear shear

$$\Sigma_t \colon (x,p) \longmapsto (x + S_t p,\ p). \tag{7.13}$$

Its trajectories are the characteristics of the transport part of (8.1). All the interaction sits in the terminal condition $p = -\mathrm{D}V_T(x + S_t p)$, an implicit equation for the momentum, solved by a sum of tree graphs (Exercise 8.1). The corresponding Hamilton–Jacobi equation $\partial_t V = \frac{1}{2}\langle \mathrm{D}V, \dot{Q}_t \mathrm{D}V\rangle$ is the Polchinski equation (7.8) without its trace term.

With noise. For a controlled diffusion $\mathrm{d}X = b(X,u)\,\mathrm{d}s + \sigma\,\mathrm{d}W$, with σ independent of the control, the stochastic maximum principle [Bis73, Pen90, YZ99] keeps the Hamiltonian and the maximisation, but the costate becomes an adapted process which solves a *backward* stochastic differential equation,

$$\mathrm{d}p_s = -\partial_x H\,\mathrm{d}s + \Upsilon_s\,\mathrm{d}W_s, \qquad p_T = -\mathrm{D}g(X_T).$$

The martingale term $\Upsilon\,\mathrm{d}W$ is the price of adaptedness: the costate may not anticipate the noise. For the field $\partial_x H = 0$, so the costate is a martingale, $p_t = \mathbb{E}_t[-\mathrm{D}V_T(X_T)]$, and since the value function is the effective potential (Theorem 7.5), $p_t = -\mathrm{D}V_t(X_t)$. The noise adds the Itô term to the Hamilton–Jacobi equation, which becomes the Polchinski equation (7.8).

In the notation of Theorem 7.5 the controlled field is $\mathrm{d}X = \dot{Q}_t^{1/2}(u\,\mathrm{d}t + \mathrm{d}W)$ and the optimal control is $u_t = \dot{Q}_t^{1/2} p_t$ (Exercise 7.3). Since p_t is the value F_t of the force, the FBSDE (7.7) is the stochastic Pontryagin system of this problem:

the field Φ is the state, and the force F is the costate.

7.3 Filtering: measuring the field

Stochastic localisation was introduced by Eldan [Eld13] to study measures in high dimension (see [Eld23] for a survey). The idea is to write a measure as a mixture of simpler ones, through a measure-valued martingale driven by noise. Proposition 7.2 says that the Polchinski process does exactly this for ν_T.

What Eldan's construction adds to the picture of Section 7.1 is a statistical reading: the process is a measurement. Assume that all the deterministic covariance operators are functions of A, hence commute, and let $K = Q_T^{-1}$, $J_t = KQ_t$, so that $J_0 = 0$, $J_T = 1$. For the heat profile, $K = Ae^{A/T^2}$ and $J_t = \mathrm{e}^{-A(t^{-2}-T^{-2})}$ is a heat semigroup, which tends to e^{-A/t^2} as $T \to \infty$. Given its terminal value, the interacting field is still the Gaussian bridge, as the density of $\mathbb{P}$ with respect to $\mathbb{P}_0$ depends on X_T only:

$$\Phi_t = J_t \Phi_T + \zeta_t, \quad \zeta_t \sim N(0, Q_t(1-J_t)), \quad \text{independent of } \Phi_T. \tag{7.14}$$

Let $\mathcal{R}_t := S_t^{-1} - Q_T^{-1}$, the **regulator**, so that $\mathcal{R}_0 = 0$ and $\dot{\mathcal{R}}_t = S_t^{-1}\dot{Q}_t S_t^{-1} \geqslant 0$. Since $S_t = Q_T(1-J_t)$, we have $S_t^{-1}J_t = \mathcal{R}_t$, and $j_t = S_t^{-1}\Phi_t$ is a noisy measurement of the terminal field, with signal-to-noise ratio the regulator:

$$j_t = \mathcal{R}_t \Phi_T + N_t, \quad N_t = S_t^{-1}\zeta_t, \quad \mathbb{E}[N_s \otimes N_t] = \mathcal{R}_{s\wedge t}, \quad N \text{ independent of } \Phi_T. \tag{7.15}$$

Let $\mathcal{K}_t(j) = \log \int \mathrm{e}^{\langle j,y\rangle - \frac{1}{2}\langle y, \mathcal{R}_t y\rangle} \nu_T(\mathrm{d}y)$. Note that $\mathcal{K}_t(S_t^{-1}x) = \frac{1}{2}\langle x, S_t^{-1}x\rangle - V_t(x)$ up to a function of t.

Proposition 7.6. (Filtering, formal; a theorem at finite cutoff) *Let $\mathcal{F}_t$ be generated by $(j_s)_{s\leqslant t}$, and let $\psi_t = \mathbb{E}_t[\Phi_T]$, $\mathrm{Cov}_t = \mathrm{Cov}(\Phi_T | \mathcal{F}_t)$. Then:*

a) The conditional law of Φ_T given $\mathcal{F}_t$ is $\mu_t^{\Phi_t}$, the Bayesian posterior, proportional to $\mathrm{e}^{\langle j_t, y\rangle - \frac{1}{2}\langle y, \mathcal{R}_t y\rangle}\nu_T(\mathrm{d}y)$. So $\psi_t = D\mathcal{K}_t(j_t)$ and $\mathrm{Cov}_t = D^2\mathcal{K}_t(j_t)$.

*b) The Brownian motion W of the FBSDE is the **innovation** of the filter: $\mathrm{d}j_t = \dot{\mathcal{R}}_t \psi_t\,\mathrm{d}t + \dot{\mathcal{R}}_t^{1/2}\,\mathrm{d}W_t$ and $\mathrm{d}\psi_t = \mathrm{Cov}_t \dot{\mathcal{R}}_t^{1/2}\,\mathrm{d}W_t$. Moreover $\psi_t = \Phi_t + S_t F_t$.*

c) The log-partition function solves the Hamilton–Jacobi equation $\partial_t \mathcal{K}_t + \frac{1}{2}\mathrm{Tr}(\dot{\mathcal{R}}_t D^2\mathcal{K}_t) + \frac{1}{2}\langle D\mathcal{K}_t, \dot{\mathcal{R}}_t D\mathcal{K}_t\rangle = 0$: this is the Polchinski equation in the variable j.

Proof. (a) Given $\Phi_T = y$, the density of the path of j with respect to that of N is $\mathrm{e}^{\langle j_t, y\rangle - \frac{1}{2}\langle y, \mathcal{R}_t y\rangle}$ (Girsanov), and Bayes' formula applies. (b) Integrate (7.6) from t to T and condition on $\mathcal{F}_t$: since F is a martingale, $\mathbb{E}_t \int_t^T \dot{Q}_s F_s\,\mathrm{d}s = S_t F_t$, so $\psi_t = \Phi_t + S_t F_t$. Differentiating $j = S^{-1}\Phi$ in (7.6), with $\partial_t S_t^{-1} = \dot{\mathcal{R}}_t$, gives the first equation. Since $\psi_t = D\mathcal{K}_t(j_t)$ is a martingale, the drift in Itô's formula vanishes and the martingale part is $D^2\mathcal{K}_t(j_t)\dot{\mathcal{R}}_t^{1/2}\,\mathrm{d}W_t$. (c) $\partial_t \mathcal{K} = -\frac{1}{2}\mathbb{E}\langle y, \dot{\mathcal{R}} y\rangle$ and $\mathrm{Tr}(\dot{\mathcal{R}} D^2\mathcal{K}) = \mathbb{E}\langle y, \dot{\mathcal{R}} y\rangle - \langle \psi, \dot{\mathcal{R}}\psi\rangle$. □

So the log-partition function of the filter is the effective potential in another coordinate, and the Polchinski flow is its Hamilton–Jacobi equation [BBD24]. The basic form of Eldan's scheme, with identity control, is the profile $S_t = (Q_T^{-1} + t)^{-1}$, $t \in [0, \infty)$, where $\mathcal{R}_t = t$ and $j_t = t\Phi_T + N_t$ with N a Brownian motion; Eldan's original construction [Eld13] uses an adapted control matrix. This is the profile of Lectures 5–6.

7.4 Where the three views meet

The three constructions give one process. Renormalisation produces it as the field at resolution t with its effective drift (Theorem 7.3), control as the optimal state (7.11), and filtering as the rescaled measurement $S_t j_t$ (Proposition 7.6). Each of them has a name for each object:

	Renormalisation	**Control**	**Filtering**
t	resolved covariance Q_t	time of the problem	information gained $\mathcal{R}_t$
Φ_t	field at resolution t	state	posterior mean for the Gaussian prior, $S_t j_t$
F_t	effective force	costate	posterior mean of the terminal force
W	fresh fluctuation of the shell	noise of the controlled field	innovation
S_t	unresolved covariance	free motion from t to T	posterior covariance for the Gaussian prior
$\psi_t = \Phi_t + S_t F_t$	mean of the conditional law $\mu_t^{\Phi_t}$	terminal state predicted by free motion	posterior mean
Cov_t	covariance of $\mu_t^{\Phi_t}$	$S_t - S_t D^2 V_t S_t$	posterior covariance
V_t	effective potential	value function	$\frac{1}{2}\langle x, S_t^{-1}x\rangle - \mathcal{K}_t(S_t^{-1}x)$, up to a function of t

In the last column, Φ_t is the estimate of the terminal field that ignores the interaction, and ψ_t is the estimate that includes it. Their difference is $S_t F_t$: the force is what the interaction adds to the estimate, in units of the free propagator S_t. The conditional covariance Cov_t replaces S_t as the propagator. Forces, propagators and covariances are the common language of the three views, and the next section writes the system in it. Only the last row of the table, and the formula for Cov_t in the control column, involve a potential.

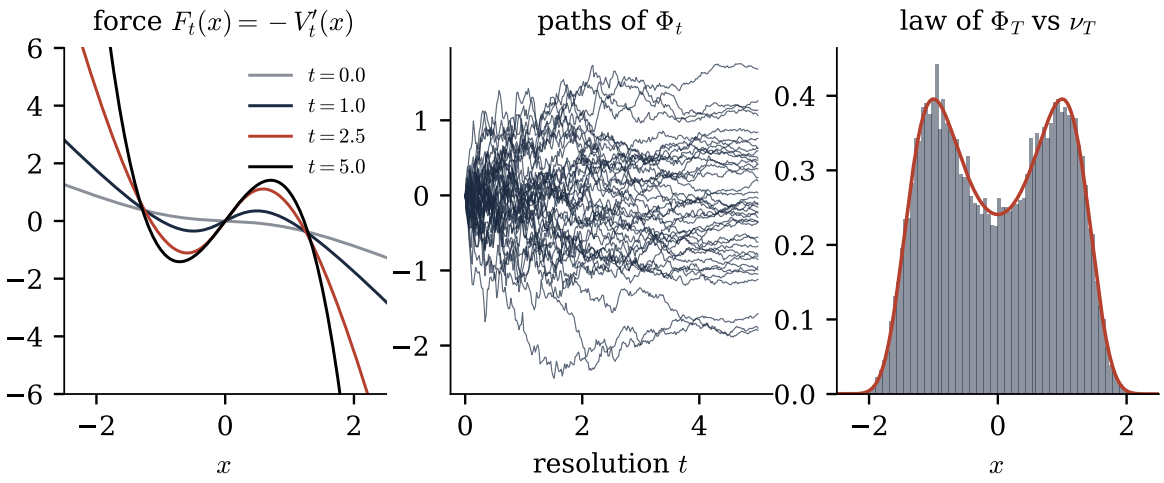


Figure 7.2. A zero-dimensional Polchinski process (Exercise 7.2): $Q_t = 1 - \mathrm{e}^{-t}$, $T = 5$, $V_T(x) = \frac{1}{2}x^4 - \frac{3}{2}x^2$. Left: the effective force $F_t = -V_t'$ is almost linear at low resolution and develops the double-well structure as $t \to T$. Centre: paths of the FBSDE (7.6), which split between the two wells (stochastic localisation). Right: histogram of Φ_T ($2 \cdot 10^4$ paths) against the density of ν_T.

Exercises

Exercise 7.1. (Quadratic interactions) Let $V_T(\varphi) = \frac{\kappa}{2}\langle\varphi,\varphi\rangle$, $\kappa > 0$. (a) Show that $V_t(\varphi) = \frac{1}{2}\langle\varphi, \kappa(1+\kappa Q_{T,t})^{-1}\varphi\rangle +$ const solves (7.8). (b) Show that ν_T is Gaussian with covariance $(Q_T^{-1} + \kappa)^{-1}$: for $Q_T \to (m^2 - \Delta)^{-1}$ this is the GFF of mass $(m^2 + \kappa)^{1/2}$. (c) Write (7.6) explicitly. It is a linear FBSDE. Solve it, and check that F_t is a martingale. *Moral:* the mass can be put either into the covariance or into the force.

Exercise 7.2. (Zero dimensions) Take $\mathcal{M}$ a point, $Q_t = q(t)$ scalar, increasing, $V_T(x) = \lambda x^4 - a x^2$ with $\lambda > 0$, $a \geqslant 0$ (a double well if $a > 0$; Figure 7.2 has $\lambda = \frac{1}{2}$, $a = \frac{3}{2}$). (a) Write (7.8) and solve it numerically for $q(t) = 1 - \mathrm{e}^{-t}$. (b) Simulate (7.6) by first computing V_t' on a grid, and compare the histogram of Φ_T with $\mathrm{e}^{-\lambda x^4 + a x^2 - x^2/2q(T)}$. (c) Compute V_t to second order in λ.

Exercise 7.3. **(Stochastic maximum principle)** (a) For the control problem (7.9), perturb an optimal control $u \to u + \varepsilon v$ and show that $p_t := \mathbb{E}_t[-\mathrm{D}V_T(X_T^u)]$ satisfies $u_t = \dot{Q}_t^{1/2} p_t$. Identify p with the force F of (7.6). (b) Show that $\mathrm{Y}_t = \mathrm{D}F_t(\Phi_t)\,\dot{Q}_t^{1/2}$.

Exercise 7.4. **(Optimal drift)** (a) Derive (7.11) from (7.9), and check that the optimal u is a martingale up to the factor $\dot{Q}^{1/2}$. (b) For the quadratic interaction of Exercise 7.1, compute the optimal drift and the value (7.9) explicitly.

Exercise 7.5. **(Cost of the leading drift)** For $V_T = \lambda \int {:}\varphi^4{:}_T$ on $\mathbb{T}_L^3$, show that

$$\frac{1}{2}\,\mathbb{E}\int_0^T \|4\lambda\,\dot{Q}_s^{1/2} {:}X_s^3{:}\|^2 \mathrm{d}s = 12\lambda^2 L^3 \int Q_T(z)^4\,\mathrm{d}z,$$

which diverges linearly in T. Relate this to the $k=4$ term of (A.6).

8 Force coordinates and potentials

The three views agree on the forward–backward system (7.7), which from now on is the primary object, with a martingale force $F_t = F_t(\Phi_t)$, $\mathrm{d}F_t = \mathrm{Y}_t\,\mathrm{d}W_t$. We look for the coordinates in which it is simplest. They are suggested by the table above: the posterior mean ψ, which we call the **forecast**, and the force. They remove the predictable part of the forward dynamics, and they integrate exactly the bilinear (tree) part of the force flow.

Sections 8.1–8.4 tell this story in terms of forces, propagators and covariances. The potential enters only through the terminal force and, for a gradient force, through the identification of the response with the conditional covariance; the system in force coordinates makes sense for a force that is not a gradient. Sections 8.5–8.7 tell it again in terms of potentials: the value function generates the canonical transformation to force coordinates, the Legendre transform connects it to the filter, and the equation for the force is the Wetterich equation for the effective average action [Wet93, BTW02].

8.1 The force and a Burgers equation

Differentiating (7.8), the effective force $F_t = -\mathrm{D}V_t$, now as a function of the field, solves

$$\partial_t F_t = -\mathrm{D}F_t[\dot{Q}_t F_t] - \frac{1}{2}\mathrm{Tr}(\dot{Q}_t \mathrm{D}^2 F_t), \qquad F_T = -\mathrm{D}V_T, \tag{8.1}$$

an infinite-dimensional, backward, viscous *Burgers equation*. The Cole–Hopf transform $F = \mathrm{D}\log u$ linearises it, which is the statement that $u = \mathrm{e}^{-V}$ solves a heat equation. Two remarks on (8.1):

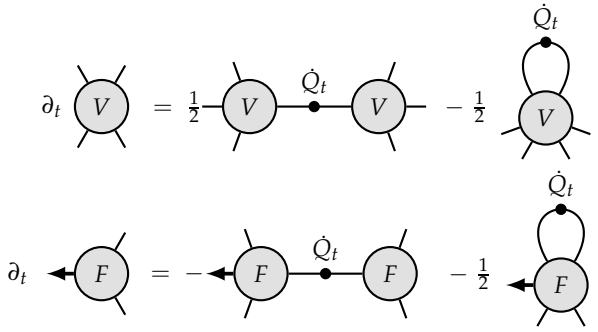


Figure 8.1. The Polchinski equation (7.8) and the Burgers equation (8.1) as graphs. A blob with n legs is the n-th derivative of V_t, or of the force F_t, whose root is the arrow; a line with a dot is the kernel $\dot{Q}_t$. The first term joins two blobs and makes a tree; the second closes a loop on one blob. Integrating from V_T and iterating generates the connected Feynman graphs, with lines $Q_{T,t}$.

i. it uses only the force, which is a local quantity (*intensive*), while V_t is *extensive*, of order of the volume. This matters in infinite volume (Section 9.4);

ii. the Burgers flows for different covariances commute, and the curvature vanishes, when $\mathrm{D}F_t$ is symmetric, i.e. when F_t is a gradient [BCG26]. For a non-gradient force, as for the general Wilson–Itô diffusions of [BCG26], the flows do not commute.

The function $F_t(\cdot)$ with $F_t = F_t(\Phi_t)$ is the *decoupling field* of the FBSDE, and the Burgers equation (8.1) is its quasilinear PDE; $\mathrm{Y}_t = \mathrm{D}F_t(\Phi_t)\,\dot{Q}_t^{1/2}$ is the martingale representation integrand of the costate. The two terms of (8.1) have distinct origins. The transport term $-\mathrm{D}F\,[\dot{Q}F]$ comes from the drift of the state, and generates the tree diagrams. The diffusion term $-\frac{1}{2}\mathrm{Tr}(\dot{Q}\,\mathrm{D}^2F)$ is the Itô correction, and generates the loops.

8.2 Removing the predictable drift

The drift $\dot{Q}_t F_t$ of the state is predictable, and the costate is a martingale. Hence the drift that remains to be accumulated on $[t,T]$ is predicted, at resolution t, by transporting the current state with the current costate along the free flow (7.13).

Proposition 8.1. (The forecast) (formal) *The conditional mean of the terminal field is*

$$\psi_t := \mathbb{E}_t[\Phi_T] = \Phi_t + S_t F_t, \tag{8.2}$$

i.e. $(\psi_t, F_t) = \Sigma_t(\Phi_t, F_t)$. *It is a martingale, with*

$$\mathrm{d}\psi_t = \dot{Q}_t^{1/2}\mathrm{d}W_t + S_t\mathrm{Y}_t\,\mathrm{d}W_t = (1 + S_t\mathrm{D}F_t(\Phi_t))\,\dot{Q}_t^{1/2}\mathrm{d}W_t. \tag{8.3}$$

Proof. The identity is in the proof of Proposition 7.6(b), and uses no commutativity. For the differential, the drift $\dot{Q}_t F_t$ of Φ cancels against $\dot{S}_t F_t$, and $S_t\,\mathrm{d}F_t = S_t\mathrm{Y}_t\,\mathrm{d}W_t$. □

So ψ is a closed observable, the *field observable* of [BCG26]. The field splits as

$$\Phi_t = \psi_t - S_t F_t\text{:} \tag{8.4}$$

a martingale, minus the drift still expected on the unresolved scales, which vanishes at $t = T$. The noise of ψ in (8.3) has two sources: the fresh fluctuation of the shell, $\dot{Q}_t^{1/2}\mathrm{d}W_t$, and the revision of the forecast of the remaining drift, $S_t\,\mathrm{d}F_t$. The coupling between the forward and the backward equations, which was in the drift of Φ, now sits in the volatility of ψ. The coordinate ψ is the *interaction picture* with respect to the noiseless (classical) part of the dynamics.

8.3 The FBSDE in the forecast coordinate

Assume that the map $x \mapsto \psi = x + S_t F_t(x)$ is invertible, and write its inverse as

$$x = \psi + S_t\,\mathcal{F}_t(\psi), \qquad \mathcal{F}_t(\psi) := -F_t(x), \tag{8.5}$$

so that $F_t = -\mathcal{F}_t(\psi_t)$: the costate is now a function of the forecast. Introduce the *response*, and recall the *regulator* of the filter:

$$H_t(\psi) := (S_t^{-1} + \mathrm{D}\mathcal{F}_t(\psi))^{-1}, \qquad \mathcal{R}_t := S_t^{-1} - Q_T^{-1}, \qquad \dot{\mathcal{R}}_t = S_t^{-1}\dot{Q}_t S_t^{-1} \geqslant 0. \tag{8.6}$$

Differentiating the two maps $x \leftrightarrow \psi$ gives $1 + S_t\mathrm{D}F_t(x) = (1 + S_t\mathrm{D}\mathcal{F}_t(\psi))^{-1} = H_t(\psi)\,S_t^{-1}$.

Theorem 8.2. (FBSDE in force coordinates) (formal) *In the coordinates* $(\psi, \mathcal{F})$, *the system* (7.7) *becomes*

$$\mathrm{d}\psi_t = H_t(\psi_t)\;S_t^{-1}\,\dot{Q}_t^{1/2}\,\mathrm{d}W_t, \qquad \mathcal{F}_t(\psi_t) = \mathbb{E}_t[\mathrm{D}V_T(\psi_T)], \tag{8.7}$$

with $\psi_0 = S_0 F_0 = \mathbb{E}[\Phi_T]$ and $\psi_T = \Phi_T$. The forward equation has no drift, and its covariance is $C_t := H_t \dot{\mathcal{R}}_t H_t^*$. The decoupling field $\mathcal{F}$ solves the pure diffusion equation

$$\partial_t \mathcal{F}_t + \tfrac{1}{2} \operatorname{Tr}(C_t \, \mathrm{D}^2 \mathcal{F}_t) = 0, \qquad \mathcal{F}_T = \mathrm{D}V_T. \tag{8.8}$$

Proof. The forward equation is (8.3) with $1 + S_t \mathrm{D}F_t = H_t S_t^{-1}$, and its covariance is

$$H_t S_t^{-1} \dot{Q}_t S_t^{-1} H_t^* = H_t \dot{\mathcal{R}}_t H_t^*.$$

The backward equation is $F_t = \mathbb{E}_t[F_T]$ with $\psi_T = \Phi_T$. Since $\mathcal{F}_t(\psi_t)$ is a martingale and ψ has no drift, the drift in Itô's formula for $\mathcal{F}_t(\psi_t)$ is the left-hand side of (8.8). □

Compare (8.8) with the Burgers equation (8.1): the transport term has been absorbed exactly by the change of coordinates, and only the Itô term is left. If it is dropped, $\mathcal{F}_t = \mathrm{D}V_T$ does not move: all the tree diagrams are contained in the coordinate change (8.5). The loops are generated only by the quadratic variation C_t of the forecast, i.e. by the randomness of the noise and of the force revisions, and the nonlinearity has moved into the field dependence of C_t. None of this uses a gradient structure or a Gaussian convolution: Theorem 8.2 holds for any FBSDE of the form (7.7) with a Markov martingale force, gradient or not, for which the coordinate change is invertible. For a general terminal force $F_T = \mathfrak{f}(\Phi_T)$, replace $\mathrm{D}V_T$ by $-\mathfrak{f}$ in (8.7) and (8.8).

8.4 Conditional moments

For the Polchinski process the response has a direct meaning.

Proposition 8.3. (formal) *If $F_t = -\mathrm{D}V_t$ then*

$$H_t(\psi_t) = \mathrm{Cov}_t = \mathrm{Cov}^{\mathbb{P}}(\Phi_T \,|\, \Phi_t) = S_t - S_t \, \mathrm{D}^2 V_t(\Phi_t) \, S_t.$$

In particular $H_t > 0$ is symmetric, $1 + S_t \mathrm{D}F_t = \mathrm{Cov}_t S_t^{-1}$ is always invertible, and $H_t \leqslant S_t$ when V_T is convex.

Proof. By (7.5), the law of Φ_T given $\mathcal{F}_t$ depends only on Φ_t, so it is also the law given Φ_t. The conditional law μ_t^{ϕ} is the tilt of $\mu^{S_t}(\cdot - \phi)$ by e^{-V_T}; differentiating twice in ϕ gives $\mathrm{Cov}(\mu_t^{\phi}) = S_t - S_t \, \mathrm{D}^2 V_t(\phi) \, S_t$. On the other hand $H_t S_t^{-1} = 1 + S_t \, \mathrm{D}F_t = 1 - S_t \, \mathrm{D}^2 V_t$. The last claim is the Brascamp–Lieb inequality. □

So the forecast and the response are the first two conditional moments of the stochastic localisation $\mu_t^{\Phi_t}$ of Proposition 7.2: $(\psi_t, \mathrm{Cov}_t) = (\mathbb{E}[\Phi_T \,|\, \Phi_t], \mathrm{Cov}(\Phi_T \,|\, \Phi_t))$ (Figure 8.2). They are tied by Itô's formula: since $\mathbb{E}_t[\Phi_T \otimes \Phi_T] = \psi_t \otimes \psi_t + \mathrm{Cov}_t$ is a martingale,

$$\mathrm{d}\langle \psi, \psi \rangle_t = C_t \, \mathrm{d}t = \mathrm{Cov}_t \dot{\mathcal{R}}_t \mathrm{Cov}_t \, \mathrm{d}t, \qquad \mathrm{d}\mathrm{Cov}_t = -\mathrm{Cov}_t \dot{\mathcal{R}}_t \mathrm{Cov}_t \, \mathrm{d}t + \mathrm{d}(\text{martingale}). \tag{8.9}$$

Conditional variance is resolved at the rate $\mathrm{Cov}_t \dot{\mathcal{R}}_t \mathrm{Cov}_t$, which for the Gaussian reference is $S_t \dot{\mathcal{R}}_t S_t = \dot{Q}_t$.

Brascamp–Lieb and Cramér–Rao bounds. The conditional law μ_t^{ϕ} has density proportional to $\mathrm{e}^{-\mathcal{U}}$, with $\mathcal{U}(y) = V_T(y) + \frac{1}{2}\langle y-\phi, S_t^{-1}(y-\phi)\rangle$ and $\mathrm{D}^2\mathcal{U} = S_t^{-1} + \mathrm{D}^2 V_T$. Two classical inequalities bound its covariance by the Hessian of $\mathcal{U}$. If $\mathrm{D}^2\mathcal{U} > 0$, the Brascamp–Lieb inequality [BL76], $\mathrm{Var}_\mu(f) \leqslant \mathbb{E}_\mu\langle \mathrm{D}f, (\mathrm{D}^2\mathcal{U})^{-1}\,\mathrm{D}f\rangle$, applied to linear functions $f = \langle h, \cdot\rangle$, and the Cramér–Rao inequality, which needs no convexity, give respectively the upper and the lower bound in

$$(\mathbb{E}_t[S_t^{-1} + \mathrm{D}^2 V_T(\Phi_T)])^{-1} \leqslant \mathrm{Cov}_t \leqslant \mathbb{E}_t[(S_t^{-1} + \mathrm{D}^2 V_T(\Phi_T))^{-1}].$$

The lower bound follows from $\mathbb{E}_\mu[(y-m)\otimes \mathrm{D}\mathcal{U}(y)] = 1$ (integration by parts) and the Cauchy–Schwarz inequality, with the Fisher information $\mathbb{E}_\mu[\mathrm{D}\mathcal{U}\otimes\mathrm{D}\mathcal{U}] = \mathbb{E}_\mu \mathrm{D}^2\mathcal{U}$ (Exercise 8.9). The two bounds coincide when V_T is quadratic, and their gap is a Jensen gap for the inverse. Through Proposition 8.3 they are two-sided bounds on the Hessian of the effective potential, i.e. on the response H_t, which is the propagator of the Wetterich equation below. Some consequences:

- If V_T is convex, then $\mathrm{Cov}_t \leqslant S_t$: the interaction can only reduce the conditional variance. Equivalently $\mathrm{D}^2 V_t \geqslant 0$: convexity is preserved along the Polchinski flow (as also follows from the Prékopa–Leindler inequality).
- If $\mathrm{D}^2 V_T \geqslant \kappa$ with $\kappa > 0$, then $\mathrm{Cov}_t \leqslant (S_t^{-1} + \kappa)^{-1}$ uniformly in the conditioning: a uniform mass in the interaction is a uniform mass in every conditional law.
- For Φ_3^4 convexity is already lost through the Wick ordering: $H_4(\varphi; c_T) = \varphi^4 - 6\,c_T\,\varphi^2 + 3\,c_T^2$, so (7.1) has curvature $-12\,\lambda\,c_T + 2\,\gamma_T$ at $\varphi = 0$, which diverges as $T \to \infty$ (this happens already for Φ_2^4). The upper bound is lost, and at low resolution the conditional law may spread over several wells (Figure 8.2). This is the large-field problem. In Lecture 5 it is handled by a clipped approximate action, whose global coercivity (Lemma 12.1 there) is a bound of Brascamp–Lieb type on the response.

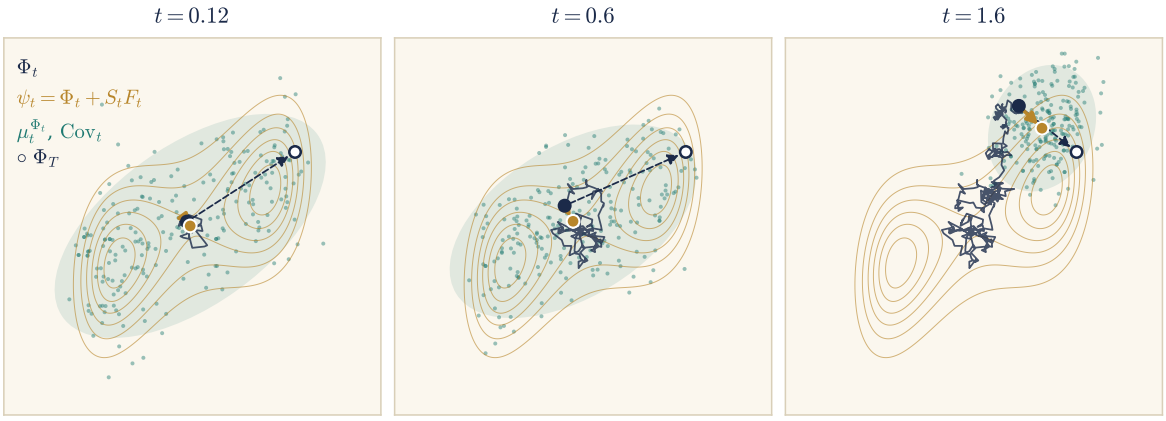


Figure 8.2. Stochastic localisation for a two-component field: $Q_t = (1-\mathrm{e}^{-t})\,1$, $T = 4$, V_T a tilted double well, and the contours show the density of ν_T. One path of (7.6) up to resolution t (navy), with its terminal value Φ_T (open circle). The forecast $\psi_t = \Phi_t + S_t F_t$ (gold) is the mean of the conditional law $\mu_t^{\Phi_t}$: teal samples, and the ellipse is two standard deviations of Cov_t. At low resolution the conditional law covers both wells; as t grows it localises on one of them.

8.5 Potentials: the value function as a generator

We now use that the force is a gradient, $F_t = -\mathrm{D}V_t$, whose graph is a Lagrangian submanifold, and that V_t is the value function of Theorem 7.5. In Hamilton–Jacobi theory a value function is more than a cost: it generates a canonical transformation. A complete family of solutions of the Hamilton–Jacobi equation generates the transformation that trivialises the dynamics. Force coordinates are this transformation.

Phase space and Lagrangian graphs. Pairs (x,p) of a field and a value of the force form the phase space $\mathbb{R}^n\times\mathbb{R}^n$, with the symplectic form $\omega=\sum_i \mathrm{d}x_i\wedge\mathrm{d}p_i$. A submanifold L of dimension n is *Lagrangian* if ω vanishes on it. The graph $L=\{(x,F(x))\}$ of a force field is Lagrangian if and only if $\sum_{i,j}\partial_j F_i\,\mathrm{d}x_i\wedge\mathrm{d}x_j=0$, i.e. $\mathrm{D}F$ is symmetric, i.e. F is (locally) a gradient: the gradient sector is the sector of Lagrangian graphs. A map that preserves ω (a *symplectic* map) sends Lagrangian submanifolds to Lagrangian submanifolds. The shear (7.13) is symplectic, since $\Sigma_t^*\,\omega=\omega+\sum_{i,j}(S_t)_{ij}\,\mathrm{d}p_j\wedge\mathrm{d}p_i=\omega$ for S_t symmetric. So it maps the graph of a gradient to a Lagrangian submanifold; if the image is again a graph over the new base variable, it is the graph of a gradient.

The free action generates the shear. The cheapest way to reach the field ψ at resolution T from the field x at resolution t, without interaction, is to minimise $\frac{1}{2}\int_t^T\|u_s\|^2\,\mathrm{d}s$ under the constraint $x+\int_t^T\dot{Q}_s^{1/2}u_s\,\mathrm{d}s=\psi$. The minimiser is $u_s=\dot{Q}_s^{1/2}p$, with the constant costate $p=S_t^{-1}(\psi-x)$, and the minimal cost is the **free action**

$$\mathcal{A}_t(x,\psi)=\frac{1}{2}\langle\psi-x,S_t^{-1}(\psi-x)\rangle. \tag{8.10}$$

The relation $\psi=x+S_tp$ is the shear (7.13). Since $D_x\mathcal{A}_t=-p$ and $D_\psi\mathcal{A}_t=p$, at fixed t

$$\mathrm{d}\mathcal{A}_t=\langle p,\,\mathrm{d}\psi\rangle-\langle p,\,\mathrm{d}x\rangle,$$

so $\mathcal{A}_t$ is the generating function of the canonical transformation Σ_t. Moreover $\partial_t\mathcal{A}_t=\frac{1}{2}\langle D_x\mathcal{A}_t,\dot{Q}_tD_x\mathcal{A}_t\rangle$: for every ψ it solves the noiseless Polchinski equation, so it is a complete integral of that equation, with parameter ψ. Jacobi's theorem then says that in the new coordinates the Hamiltonian H_* vanishes: ψ and p are constants of motion.

This is what force coordinates are:

(ψ_t,F_t) are the Jacobi coordinates of the free Hamiltonian of the field.

The noise does not destroy this: the constants of motion become martingales, ψ_t by Proposition 8.1 and F_t by Theorem 7.3. The only dynamics left is their quadratic variation.

The value function generates the graph. The graph of the force is Lagrangian, hence so is its image under the shear. The image is a graph over ψ: the map $x\mapsto x+S_tF_t(x)$ is S_t times the gradient of $\frac{1}{2}\langle x,S_t^{-1}x\rangle-V_t(x)$, which is strictly convex since its Hessian is $S_t^{-1}\mathrm{Cov}_tS_t^{-1}>0$ (Proposition 8.3), so the map is injective. Hence the image is the graph of a gradient:

$$\mathcal{F}_t=\mathrm{D}\Gamma_t,\qquad H_t=(S_t^{-1}+\mathrm{D}^2\Gamma_t)^{-1},\qquad \Gamma_T=V_T. \tag{8.11}$$

These two generating functions are related by the Moreau (infimal convolution) transform

$$V_t(x)=\inf_\psi\Big\{\Gamma_t(\psi)+\tfrac{1}{2}\langle x-\psi,S_t^{-1}(x-\psi)\rangle\Big\}, \tag{8.12}$$

whose optimality conditions $x=\psi+S_t\,\mathrm{D}\Gamma_t(\psi)$, $\mathrm{D}V_t(x)=\mathrm{D}\Gamma_t(\psi)$ are (8.5). It is well defined by the same convexity: up to the change of variable $j=S_t^{-1}x$, it is a Legendre transform. We use (8.12) to fix the additive constant in Γ_t. Figure 8.3 shows the construction in zero dimensions.

At the minimiser, $V_t(x)=\Gamma_t(\psi)+\mathcal{A}_t(x,\psi)$: the value function is the cost Γ_t attached to the forecast, plus the cost of reaching the forecast by free motion. This is the rule for transporting a generating function along a canonical transformation: $\mathrm{d}V_t=\langle D\Gamma_t,\,\mathrm{d}\psi\rangle+\langle p,\,\mathrm{d}\psi\rangle-\langle p,\,\mathrm{d}x\rangle=-\langle F_t,\,\mathrm{d}x\rangle$, since $D\Gamma_t(\psi)=-F_t=-p$.

No convexity is needed. The construction uses no convexity of V_T, V_t or Γ_t. The convex function is $g_t(x) = \frac{1}{2}\langle x, S_t^{-1}x\rangle - V_t(x)$: with $j = S_t^{-1}x$ it is, up to a constant, the log-Laplace transform of the finite measure $e^{-V_T(y)-\frac{1}{2}\langle y,S_t^{-1}y\rangle}\,dy$, for every V_T bounded below. Its gradient in j is the conditional mean ψ, and it is strictly convex because $\mathrm{Cov}_t > 0$. In finite dimensions, when V_T is finite, this exponential family is regular with full support, so $j \mapsto \psi$ is a bijection of $\mathbb{R}^n$: the image of the shear is a graph over the whole ψ space, also when V_t is a double well.

For Γ_t one only gets semiconvexity, $S_t^{-1} + D^2\Gamma_t = \mathrm{Cov}_t^{-1} > 0$, i.e. $\Gamma_t + \frac{1}{2}\langle\cdot, S_t^{-1}\cdot\rangle$ is strictly convex, which is what (8.12) requires. At $t = 0$, where $S_0 = Q_T$, this semiconvexity is the convexity of the effective average action of Section 8.6, Γ_0^{EAA}. Γ_t itself is convex if and only if $\mathrm{Cov}_t \leqslant S_t$, which holds for convex V_T but may fail for a double well at low resolution. Convexity matters only without noise: the tree-level potential (8.17) is merely semiconcave, and unless $S_t^{-1} + D^2V_T > 0$ the characteristics cross, producing caustics and shocks of the force. The noise removes them.

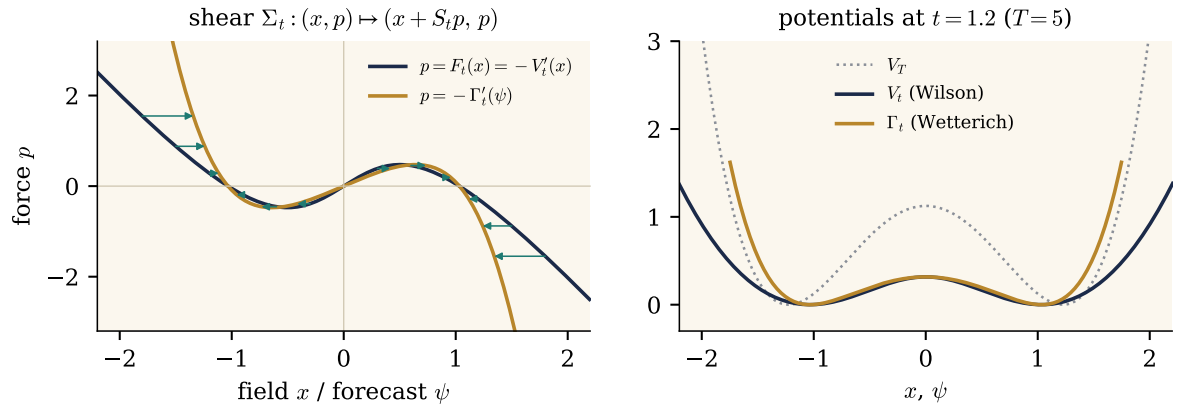


Figure 8.3. The zero-dimensional model of Figure 7.2 at $t = 1.2$. Left: the shear Σ_t (teal arrows) maps the graph of the force $F_t = -V_t'$ (navy) to a graph over the forecast ψ (gold), which is the graph of $-\Gamma_t'$. Right: the bare potential V_T, the effective potential V_t and Γ_t, each shifted to have minimum 0.

8.6 The Legendre transform and Wetterich's equation

Theorem 8.4. (Wetterich equation [Wet93]) (formal) Γ_t *solves*

$$\partial_t\Gamma_t = \tfrac{1}{2}\mathrm{Tr}[\dot{\mathcal{R}}_t\,((S_t^{-1} + D^2\Gamma_t)^{-1} - S_t)], \qquad \Gamma_T = V_T. \tag{8.13}$$

Equivalently, the effective average action $\Gamma_t^{\mathrm{EAA}}(\psi) := \Gamma_t(\psi) + \frac{1}{2}\langle\psi, Q_T^{-1}\psi\rangle$ *solves, up to a field-independent term,*

$$\partial_t\Gamma_t^{\mathrm{EAA}} = \tfrac{1}{2}\mathrm{Tr}[\dot{\mathcal{R}}_t\,(D^2\Gamma_t^{\mathrm{EAA}} + \mathcal{R}_t)^{-1}]. \tag{8.14}$$

It interpolates between the bare action $\Gamma_T^{\mathrm{EAA}} = \frac{1}{2}\langle\psi, Q_T^{-1}\psi\rangle + V_T(\psi)$ $(\mathcal{R}_T = \infty)$ *and the 1PI effective action* Γ_0^{EAA} *of* ν_T $(\mathcal{R}_0 = 0)$.

Proof. Integrate (8.8) in field space. With $\mathcal{F} = D\Gamma$, the derivative of $H = (S^{-1} + D^2\Gamma)^{-1}$ in a direction v is $-H\,D^3\Gamma[v]\,H$, so by the total symmetry of $D^3\Gamma$,

$$D_v\mathrm{Tr}(\dot{\mathcal{R}}_t H_t) = -\mathrm{Tr}(H_t\,\dot{\mathcal{R}}_t H_t\,D^3\Gamma_t[v]) = -\mathrm{Tr}(C_t\,D^2\mathcal{F}_t)[v].$$

Hence (8.8) reads $\mathrm{D}\big(\partial_t\Gamma_t-\frac{1}{2}\mathrm{Tr}(\dot{\mathcal{R}}_t H_t)\big)=0$, and (8.13) holds up to a function of t alone. That this function is $-\frac{1}{2}\mathrm{Tr}(\dot{\mathcal{R}}_t S_t)$ with the normalisation (8.12) follows from the Polchinski equation by the envelope theorem (Exercise 8.4). For (8.14), note that $\mathrm{D}^2\Gamma_t^{\mathrm{EAA}}+\mathcal{R}_t=\mathrm{D}^2\Gamma_t+S_t^{-1}$. For the endpoints: $S_T=0$; and at $t=0$, $S_0=Q_T$, so (8.12) identifies $\frac{1}{2}\langle x,Q_T^{-1}x\rangle-V_0(x)$, up to a constant, with the cumulant generating function of ν_T at $Q_T^{-1}x$, whose Legendre transform is Γ_0^{EAA} (Exercise 8.6). □

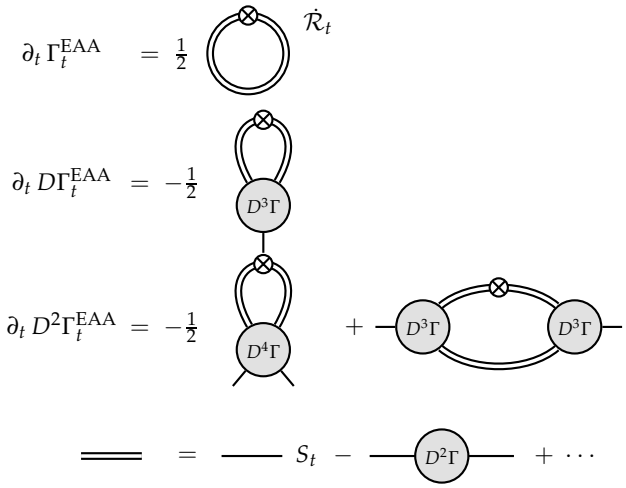


Figure 8.4. The Wetterich equation (8.14) as graphs. The double line is the full response $(D^2\Gamma_t^{\mathrm{EAA}}+\mathcal{R}_t)^{-1}=(S_t^{-1}+D^2\Gamma_t)^{-1}$ and the crossed circle is one insertion of $\dot{\mathcal{R}}_t$. There is exactly one loop: the tree term of Figure 8.1 has been absorbed by the Legendre transform. Each field derivative brings down a higher vertex, so the hierarchy does not close at a finite order. The bottom row expands the response around S_t.

Thus the Wetterich equation is the gradient-integrated form of the force equation (8.8), i.e. of the backward half of the FBSDE in forecast coordinates [Mor94, BCG26]. Every object in it is an observable of the Polchinski process:

Wetterich	FBSDE
field argument ψ	forecast $\psi_t=\mathbb{E}_t[\Phi_T]$, the state transported by Σ_t
$\mathrm{D}\Gamma_t(\psi_t)$	minus the costate, $-F_t=\mathbb{E}_t[\mathrm{D}V_T(\Phi_T)]$
propagator $(\mathrm{D}^2\Gamma_t^{\mathrm{EAA}}+\mathcal{R}_t)^{-1}$	conditional covariance Cov_t
regulator $\mathcal{R}_t=S_t^{-1}-Q_T^{-1}$	precision of the unresolved shell, minus that of the prior
$\partial_t\Gamma_t$	$\frac{1}{2}\mathrm{Tr}[\dot{\mathcal{R}}_t(\mathrm{Cov}_t-S_t)]$

The flow of Γ is driven by the difference between the interacting and the free conditional covariances, paired with the shell $\dot{\mathcal{R}}_t$. It is *one-loop exact*, with the full propagator inside the trace. In the Gaussian case $\mathrm{D}^2\Gamma_t$ is constant and only the constant term of Γ flows. For the heat profile with $T=\infty$, $S_t=C\,(1-\mathrm{e}^{-A/t^2})$ and

$$\mathcal{R}_t=\frac{A}{\mathrm{e}^{A/t^2}-1}, \tag{8.15}$$

which behaves like a mass term t^2 on the modes $|k|\ll t$ (for $m\ll t$) and is exponentially small on the modes $|k|\gg t$: the exponential regulator of the functional renormalisation group (Figure 8.5).

Proposition 8.5. (The Legendre transform, formal; a theorem at finite cutoff) *In the setting of Proposition 7.6, the Legendre transform of the log-partition function of the filter gives the effective action: $\mathcal{K}_t^*(\psi)=\Gamma_t(\psi)+\frac{1}{2}\langle\psi,S_t^{-1}\psi\rangle=\Gamma_t^{\mathrm{EAA}}(\psi)+\frac{1}{2}\langle\psi,\mathcal{R}_t\psi\rangle$ up to a function of t, and differentiating at fixed ψ gives the Wetterich equation (8.13) again.*

Proof. By (8.12) and the formula for $\mathscr{K}$. At fixed ψ, with $j = D\mathscr{K}^*(\psi)$, the envelope theorem and Proposition 7.6(c) give $\partial_t \mathscr{K}^* = \frac{1}{2}\mathrm{Tr}(\dot{\mathscr{R}}\mathrm{Cov}) + \frac{1}{2}\langle\psi, \dot{\mathscr{R}}\psi\rangle$, and $\partial_t \frac{1}{2}\langle\psi, S^{-1}\psi\rangle = \frac{1}{2}\langle\psi, \dot{\mathscr{R}}\psi\rangle$. Hence $\partial_t \Gamma_t = \frac{1}{2}\mathrm{Tr}(\dot{\mathscr{R}}\mathrm{Cov})$ up to a function of t, which is $-\frac{1}{2}\mathrm{Tr}(\dot{\mathscr{R}}S)$ with the normalisation (8.12). □

So one Lagrangian graph has three generators: V_t in the field x, $\mathscr{K}_t$ in the measurement $j = S_t^{-1}x$, and Γ_t in the forecast ψ. The Polchinski flow is the Hamilton–Jacobi equation of the log-partition function of the localisation; the Wetterich flow is its Legendre dual [Mor94], whose argument is the posterior mean, and whose regulator is the information gained. At $t = 0$, where $\mathscr{R}_0 = 0$, $\mathscr{K}_0$ is the generating function Λ of the cumulants of ν_T below.

Probabilistic meaning of Γ. At $t = 0$ the effective average action is the 1PI effective action of ν_T,

$$\Gamma_0^{\mathrm{EAA}}(\psi) = V_0(0) + \sup_j \{\langle j, \psi\rangle - \Lambda(j)\}, \qquad \Lambda(j) = \log \int \mathrm{e}^{\langle j, \varphi\rangle}\, \nu_T(\mathrm{d}\varphi),$$

the generating functional of the vertex functions introduced by Jona-Lasinio [JL64]. It is a probabilistic object, as Jona-Lasinio emphasised in his probabilistic reading of the renormalisation group [JL75, JL01]; for the link with large deviations see Section VIII of [JL01]. By Cramér's theorem the empirical mean of N independent samples of ν_T takes the value ψ with probability $\approx \exp\left(-N\left(\Gamma_0^{\mathrm{EAA}}(\psi) - V_0(0)\right)\right)$, so $\Gamma_0^{\mathrm{EAA}} - V_0(0)$ is the rate function of the large deviations of averages of the field. In particular Γ_0^{EAA} is convex even when V_T is not, it is minimal at the mean of ν_T, and its Hessian there is the inverse covariance. Formally the same holds for the field averaged over a large volume, with the volume as the large parameter and the effective potential of quantum field theory (Γ per unit volume at constant ψ, not V_t) as rate function; in that limit flat parts may appear (the Maxwell construction).

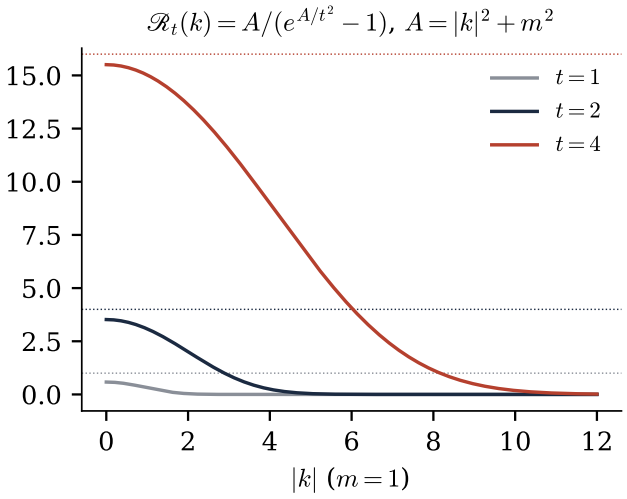


Figure 8.5. The regulator $\mathscr{R}_t$ of the heat profile ($T = \infty$, $m = 1$): close to the mass term t^2 (dotted) on the modes $|k| \ll t$, exponentially small on the modes $|k| \gg t$.

8.7 Variational meaning and the tree level

Constrained free energies. (formal) By the Gibbs variational principle applied to the conditional law (7.5), and the identity $\mathrm{Ent}(\nu|\mu^S(\cdot - x)) = \mathrm{Ent}(\nu|\mu^S(\cdot - \psi)) + \frac{1}{2}\langle x - \psi, S^{-1}(x - \psi)\rangle$ for ν of mean ψ,

$$\Gamma_t(\psi) = \inf\Big\{\int V_T\,\mathrm{d}\nu + \mathrm{Ent}(\nu\,|\,\mu^{S_t}(\cdot - \psi))\colon \int \varphi\ \nu(\mathrm{d}\varphi) = \psi\Big\}. \tag{8.16}$$

So Γ_t, the effective average action up to the quadratic term $\frac{1}{2}\langle\psi, Q_T^{-1}\psi\rangle$, is the free energy of the unresolved shell constrained to the forecast ψ, and (8.12) is the minimisation over the forecast. In the language of Theorem 7.5: V_t is the value function of the control problem, and Γ_t is its Moreau transform, i.e. the value function expressed in the forecast coordinate, the interaction picture of the classical part of the dynamics.

The tree level. Dropping the Itô term in (8.8) gives $\Gamma_t = V_T$, and then (8.12) becomes the Hopf–Lax formula

$$V_t^{\mathrm{tree}}(x) = \inf_y \Big\{ V_T(y) + \tfrac{1}{2}\langle y-x, S_t^{-1}(y-x)\rangle \Big\}, \tag{8.17}$$

the solution of the noiseless Polchinski equation $\partial_t V = \frac{1}{2}\langle \mathrm{D}V, \dot{Q}\,\mathrm{D}V\rangle$. It is the saddle point of the Gaussian convolution (7.3). Its minimiser solves the classical field equation $S_t^{-1}(y-x) + \mathrm{D}V_T(y) = 0$ in the shell $[t,T]$ with background x, and expanding it in V_T produces exactly the tree diagrams.

Exercises

Exercise 8.1. (Characteristics) (a) Show that the transport equation $\partial_t F + \mathrm{D}F\,[\dot{Q}F] = 0$ is solved by $F_t(x) = F_T(y)$, $y = x + S_t F_T(y)$, and that its characteristics are the trajectories of (7.13). (b) Verify that (8.17) solves $\partial_t V = \frac{1}{2}\langle \mathrm{D}V, \dot{Q}\,\mathrm{D}V\rangle$, and that it is the limit $\hbar \to 0$ of $-\hbar \log \int \mathrm{e}^{-V_T(x+\zeta)/\hbar}\mu^{\hbar S_t}(\mathrm{d}\zeta)$. (c) For $V_T(y) = \lambda y^4$ in zero dimensions, solve the equation for y as a power series in λ, and identify the coefficients as tree graphs.

Exercise 8.2. (Conditional moments) (a) Prove $\mathrm{Cov}(\mu_t^{\phi}) = S_t - S_t\,\mathrm{D}^2V_t(\phi)\,S_t$. (b) Prove (8.9).

Exercise 8.3. (A symplectic shear) (a) Show that Σ_t preserves $\sum_i \mathrm{d}x_i \wedge \mathrm{d}p_i$ iff S_t is symmetric. (b) Show that the image of the graph of $-\mathrm{D}V_t$ under Σ_t is the graph of $-\mathrm{D}\Gamma_t$, with Γ_t given by (8.12). (c) For a non-gradient force, show that Theorem 8.2 still holds, but that $\mathcal{F}_t$ need not be a gradient.

Exercise 8.4. (The constant) Differentiate (8.12) in t at fixed x (envelope theorem), and use (7.8) to show that the quadratic terms cancel and $\partial_t \Gamma_t = -\frac{1}{2}\mathrm{Tr}(\dot{Q}_t\,\mathrm{D}^2V_t)$. Deduce (8.13) with the stated constant.

Exercise 8.5. (Gaussian check) For $V_T(\varphi) = \frac{\kappa}{2}\langle\varphi,\varphi\rangle$, show that $\Gamma_t(\psi) = \frac{\kappa}{2}\langle\psi,\psi\rangle + \mathrm{const}$, and solve the FBSDE (8.7) explicitly. Compare with Exercise 7.1.

Exercise 8.6. (1PI effective action) (a) Let $\mathcal{K}(j) = \log \int \mathrm{e}^{\langle j,\varphi\rangle}\mathrm{e}^{-V_T(\varphi)}\mu^{Q_T}(\mathrm{d}\varphi)$. Show that $\Gamma_0^{\mathrm{EAA}} = \mathcal{K}^*$, and that $\mathrm{D}^2\Gamma_0^{\mathrm{EAA}}(\psi)$ is the inverse covariance of the tilted measure with mean ψ. (b) Integrate (8.13) with $\mathrm{D}^2\Gamma_t$ replaced by D^2V_T, and obtain the one-loop formula $\Gamma_t \approx V_T + \frac{1}{2}\mathrm{Trlog}\,(1 + S_t\,\mathrm{D}^2V_T)$.

Exercise 8.7. (Trees versus loops) Using (8.12), show that $V_t = \Gamma_t - \frac{1}{2}\langle \mathrm{D}\Gamma_t, S_t\,\mathrm{D}\Gamma_t\rangle + O(\lambda^3)$ for the Φ^4 interaction (7.1). Deduce from Proposition A.2 that, to order λ^2, Γ_t is obtained from V_t by removing exactly the term $k=1$, the one-particle-reducible tree; the bubble ($k=2$), the sunset ($k=3$) and the vacuum term ($k=4$) are 1PI and remain. Where does the sunset come from in (8.13)?

Exercise 8.8. (Regulators) Derive (8.15). Compute $\mathcal{R}_t$ for the sharp Fourier profile, and for $\hat{Q}_t(k) = (|k|^2 + m^2)^{-1} - (|k|^2 + m^2 + t^2)^{-1}$ (whose kernels are not smooth, but the formulas make sense). In the last case show that $\mathcal{R}_t = t^2$ when $T = \infty$: a mass regulator.

Exercise 8.9. (Brascamp–Lieb and Cramér–Rao) Let $\mu(\mathrm{d}y) \propto \mathrm{e}^{-U(y)}\,\mathrm{d}y$ on $\mathbb{R}^n$, with mean m. (a) Show that $\mathbb{E}_\mu[(y-m)\otimes \mathrm{D}U] = 1$ and $\mathbb{E}_\mu[\mathrm{D}U \otimes \mathrm{D}U] = \mathbb{E}_\mu \mathrm{D}^2U$, and deduce $\mathrm{Cov}_\mu \geqslant (\mathbb{E}_\mu \mathrm{D}^2U)^{-1}$. (b) For $\mathrm{D}^2U > 0$, prove the Brascamp–Lieb inequality for linear f by Hörmander's L^2 method: solve $Lg = f - \mathbb{E}_\mu f$ with $L = -\Delta + \langle \mathrm{D}U, D\rangle$ and use $\mathbb{E}_\mu[(Lg)^2] \geqslant \mathbb{E}_\mu\langle \mathrm{D}g, \mathrm{D}^2U\,\mathrm{D}g\rangle$. (c) In zero dimensions, with $S_t = S$ and $V_T(y) = \lambda y^4$, compute both bounds numerically and compare with the conditional variance; repeat for the double well $V_T(y) = \frac{1}{2}y^4 - \frac{3}{2}y^2$.

Exercise 8.10. (The free action) (a) Prove (8.10), and that $\partial_t \mathcal{A}_t = \frac{1}{2}\langle D_x\mathcal{A}_t, \dot{Q}_t D_x\mathcal{A}_t\rangle$. (b) From the Moreau formula (8.12), recover the relations (8.5) by differentiating $V_t(x) = \Gamma_t(\psi) + \mathcal{A}_t(x,\psi)$.

9 Observables and field equations

In Lecture 3 the primary objects were the stochastic observables, martingales in the filtration of finite-resolution measurements [BCG26]. The filtering point of view makes this natural: an observable is a posterior mean $\mathbb{E}_t[G(\Phi_T)]$. We have already met the force F_t and the forecast ψ_t, and the field Φ_t is not one of them. This section develops the language for the interacting theory, and then draws the main consequence of the filter: a field equation at every resolution, with a noise whose moments are controlled a priori.

9.1 Observables of the interacting theory

For a functional G of the terminal field, the observable $O_t = \mathbb{E}_t^{\mathbb{P}}[G(\Phi_T)]$ is $O_t = G_t(\Phi_t)$, with

$$G_t(\phi) = \mu_t^{\phi}(G) = \mathrm{e}^{V_t(\phi)} \int G(\phi+\zeta)\, \mathrm{e}^{-V_T(\phi+\zeta)} \mu^{Q_{T,t}}(\mathrm{d}\zeta). \tag{9.1}$$

Formally, G_t solves the backward Kolmogorov equation of the interacting dynamics,

$$\partial_t G_t + \tfrac{1}{2}\mathrm{Tr}(\dot{Q}_t \mathrm{D}^2 G_t) - \langle \dot{Q}_t \mathrm{D}V_t, \mathrm{D}G_t\rangle = 0, \qquad G_T = G, \tag{9.2}$$

which is the Doob transform of the Gaussian heat equation by e^{-V_t} (Exercise 9.1). Conversely, any solution of (9.2) yields an observable $G_t(\Phi_t)$; at finite T it is closed, but as $T \to \infty$ it need not be. The force is the case $G = -\mathrm{D}V_T$, and is special because it also drives the dynamics.

Tangent observables. Let V_T^{θ} depend on a coupling θ, and let $U_t := \partial_\theta V_t^{\theta}$. Differentiating (7.3) gives $U_t(\Phi_t) = \mathbb{E}_t^{\theta}[\partial_\theta V_T^{\theta}(\Phi_T)]$, an observable, and

$$\partial_\theta \mathbb{E}^{\nu_T^{\theta}}[G] = -\mathrm{Cov}^{\nu_T^{\theta}}(G, \partial_\theta V_T^{\theta}). \tag{9.3}$$

Thus the tangent observables, such as the running $:\varphi^4:$ for $\theta = \lambda$ (together with the derivative of the counterterm $\partial_\lambda \gamma_T \int :\varphi^2:$), describe the response of the theory to a change of coupling. They are the observables conjugate to the coordinates of theory space.

9.2 The field equation at every resolution

The measurement noise N_t of (7.15) is not observed, but its posterior mean is: from $j_t = \mathscr{R}_t \Phi_T + N_t$ we get $\mathbb{E}_t N_t = j_t - \mathscr{R}_t \psi_t$, which is the **noise observable** η_t below. This gives a field equation at every resolution.

Theorem 9.1. (DS equations at resolution t) (formal) *The* noise observable

$$\eta_t := K\,\Phi_t - J_t F_t \tag{9.4}$$

is a centred $\mathbb{P}$-martingale, with $\mathrm{d}\eta_t = K\dot{Q}_t^{1/2}\mathrm{d}W_t - J_t\,\mathrm{d}F_t$. For every $t \leqslant T$, every h and every functional G,

$$\mathbb{E}^{\mathbb{P}}[D_{J_t h} G(\Phi_t)] = \mathbb{E}^{\mathbb{P}}[G(\Phi_t)\,\langle h, \eta_t\rangle] = \mathbb{E}^{\mathbb{P}}[G(\Phi_t)\,\langle h, K\,\Phi_t - J_t F_t\rangle]. \tag{9.5}$$

At $t = T$ this is the Dyson–Schwinger equation of ν_T,

$$\mathbb{E}^{\nu_T}[D_h G] = \mathbb{E}^{\nu_T}[G\,\langle h, K\varphi + \mathrm{D}V_T(\varphi)\rangle]. \tag{9.6}$$

Proof. From (7.6) and $\dot{J}_t = K\dot{Q}_t$, $\mathrm{d}\eta_t = K(\dot{Q}_t F_t \mathrm{d}t + \dot{Q}_t^{1/2}\mathrm{d}W_t) - K\dot{Q}_t F_t \mathrm{d}t - J_t \mathrm{d}F_t$. The drift cancels, F is a martingale and $\eta_0 = 0$. For (9.5) use Proposition 7.2: $\Phi_t \sim \nu_t \propto \mathrm{e}^{-V_t}\mu^{Q_t}$. Gaussian integration by parts for μ^{Q_t} in the direction $Q_t k$ gives $\mathbb{E}^{\nu_t}[D_{Q_t k} G] = \mathbb{E}^{\nu_t}[G\,(\langle k, \varphi\rangle + \langle Q_t k, \mathrm{D}V_t(\varphi)\rangle)]$. Take $k = Kh$, so that $Q_t k = J_t h$, and recall $F_t = -\mathrm{D}V_t(\Phi_t)$. □

Formally letting $T\to\infty$, $K\to A$, and (9.4) becomes an exact identity between observables at every resolution:

$$A\,\Phi_t = A\,Q_t\,F_t + \eta_t, \tag{9.7}$$

with $A\,Q_t = \mathrm{e}^{-A/t^2}$ for the heat profile. This is the Euclidean field equation $A\varphi = -V'(\varphi) + \text{noise}$ of Lectures 1–2, seen at resolution t. It needs no products of distributions: the nonlinearity enters only through the force *observable* F_t, smeared at scale $1/t$. For the GFF ($F=0$), (9.5) is Gaussian integration by parts for μ^{Q_t}. For Φ^4_3, where F_t plays the role of the renormalised cube, the limit $t\to\infty$ of (9.7) corresponds to the renormalised DS equation of [GH21]. Lecture 6 proves the field equation at every resolution for the limit (Theorem 18.9).

Remark 9.2. (Comparison with stochastic quantisation) In parabolic or elliptic stochastic quantisation (Lecture 3, §4) the field equation is imposed in an auxiliary time, and the field at fixed time has the right law. Here the field equation (9.7) holds at each resolution, and the auxiliary parameter is the resolution itself. Taking conditional expectations of an SPDE solution with respect to a scale filtration gives identities of the same shape, but with a different noise covariance: the martingale $J_t\,\mathrm{d}F_t$ in $\mathrm{d}\eta_t$ is essential [BCG26].

Corollary 9.3. (Dyson–Schwinger along the scales, formal) *Let $t_1\leqslant\ldots\leqslant t_k\leqslant t<T$, $h,h_1,\ldots,h_k$ be test functions and $G=g(\langle\Phi_{t_1},h_1\rangle,\ldots,\langle\Phi_{t_k},h_k\rangle)$. Then*

$$\mathbb{E}[G\langle\eta_t,h\rangle] = \mathbb{E}\sum_{i=1}^{k}\partial_i g\langle J_{t_i}h,h_i\rangle, \tag{9.8}$$

that is, the derivative of G in the direction which shifts the field at each scale s by $J_s h$. The increments are consistent: $\mathbb{E}[G\langle\eta_t-\eta_s,h\rangle]=0$ for $t_k\leqslant s\leqslant t$.

Proof. G is $\mathcal{F}_t$-measurable, so $\mathbb{E}[G\langle\eta_t,h\rangle]=\mathbb{E}[G\langle N_t,h\rangle]$. By (7.14), $\Phi_s = J_s\Phi_T + S_sN_s$, and for $s\leqslant t$ the covariance of $\langle N_t,h\rangle$ and $\langle S_sN_s,k\rangle$ is $\langle S_s\mathcal{R}_s h,k\rangle=\langle J_s h,k\rangle$. Apply Gaussian integration by parts in N, given Φ_T. The last claim is the martingale property of η. □

Remark 9.4. (Closed versus local) At finite T the force is a closed observable, $F_t=\mathbb{E}_t[F_T]$. For Φ^4, $F_T(x) = -4\lambda H_3(\Phi_T(x);c_T) - 2\gamma_T H_1(\Phi_T(x);c_T)$ is local. As $T\to\infty$ one expects the force to converge as an observable. In $d=2$ the limit is closed, $F_\infty = -4\lambda{:}\Phi^3_\infty{:}$. In $d=3$ it is expected not to be. If it were, $F_t=\mathbb{E}_t[F_\infty]$ with $F_\infty\in L^1$, the DS equation of Theorem 9.1 would pass to the limit ($T\to\infty$, then $t\to\infty$), with F_∞ paired only with functionals of Φ_∞. Conditioning on $\sigma(\Phi_\infty)$ would then give a logarithmic derivative of the Φ^4_3 measure that is an integrable function of the field, $\langle Ah,\Phi_\infty\rangle - \mathbb{E}[F_\infty(h)|\Phi_\infty]$. This is not expected: the regularised logarithmic derivatives do not converge in L^2 [ALZ06]. The limiting force would then be an observable in the proper sense, a compatible family with no terminal representative. Locality of the limiting force is the crux of the construction [BCG26].

9.3 Gaussian bounds for the noise

The noise observable η_t is not Gaussian: it depends on the unknown law of the interacting field. But the filter says what it is, the conditional expectation of the Gaussian noise N_t, so its moments are bounded by those of an explicit Gaussian, uniformly in the interaction.

Work at finite cutoff, where Φ_t has a Lebesgue density p_t, and let $0<t<T$. By Proposition 7.2, $-\mathrm{Dlog}\,p_t(a) = Q_t^{-1}a - F_t(a)$, and since $J_tQ_t^{-1}=K$,

$$\eta_t = -J_t\,\mathrm{Dlog}\,p_t(\Phi_t). \tag{9.9}$$

The noise is the score of the law of the field at resolution t, seen through J_t.

Only deterministic covariance operators, all functions of A, are commuted below. The variance that appears is the regulator (8.6) of the Wetterich equation: since $S_t=Q_T(1-J_t)$,

$$J_t^2\,(Q_t(1-J_t))^{-1} = KJ_t(1-J_t)^{-1} = S_t^{-1} - Q_T^{-1} = \mathcal{R}_t. \tag{9.10}$$

Proposition 9.5. (Sub-Gaussian noise) *Let $0<t<T$. Then $\eta_t=\mathbb{E}[J_t(Q_t(1-J_t))^{-1}\zeta_t|\Phi_t]$, and η_t is dominated in the convex order by $N(0,\mathcal{R}_t)$: $\mathbb{E}\,\Psi(\eta_t)\leqslant\mathbb{E}\,\Psi(\Xi)$, $\Xi\sim N(0,\mathcal{R}_t)$, for every convex Ψ for which both sides are defined. In particular, for every h and $u\in\mathbb{R}$,*

$$\mathbb{E}\,\mathrm{e}^{u\langle h,\eta_t\rangle}\leqslant\exp\left(\frac{u^2}{2}\langle h,\mathcal{R}_t h\rangle\right): \tag{9.11}$$

every smeared noise observable is centred and sub-Gaussian, with the regulator as variance and with no convexity assumption on the interaction.

Proof. Write $B=Q_t(1-J_t)$ and g for the density of $N(0,B)$. Then $p_t(a)=\mathbb{E}\,g(a-J_t\Phi_T)$ and $\mathrm{D}g(z)=-B^{-1}z\,g(z)$, so $-\mathrm{Dlog}\,p_t(\Phi_t)=\mathbb{E}[B^{-1}\zeta_t|\Phi_t]$; apply J_t and use (9.9). By conditional Jensen, $\Psi(\eta_t)\leqslant\mathbb{E}[\Psi(J_tB^{-1}\zeta_t)|\Phi_t]$, and $J_tB^{-1}\zeta_t$ is centred Gaussian with covariance $J_tB^{-1}J_t=\mathcal{R}_t$. □

For the heat profile (compare (8.15)), $K=A\,\mathrm{e}^{A/T^2}$ and $J_t=\mathrm{e}^{-A(t^{-2}-T^{-2})}$ give

$$\mathcal{R}_t=\frac{A\,\mathrm{e}^{A/T^2}}{\mathrm{e}^{A(t^{-2}-T^{-2})}-1}\leqslant\frac{4}{3}t^2\quad(T\geqslant 2t),\qquad\mathcal{R}_t=\frac{A}{\mathrm{e}^{A/t^2}-1}\leqslant t^2\quad(T=\infty)$$

(Exercise 9.5). At a finite spatial cutoff all of this holds with terminal resolution $T=\infty$, before any continuum measure is constructed.

The roughness of the noise. Compare spectral densities for the heat profile with $T=\infty$ and $A=|k|^2+m^2$: white noise ξ has density 1, the GFF has A^{-1}, the free noise $\eta_t=AX_t$ has $A\,\mathrm{e}^{-A/t^2}$, and the bound of Proposition 9.5 is

$$\mathcal{R}_t(k)=\frac{A}{\mathrm{e}^{A/t^2}-1}\approx\begin{cases}t^2, & |k|\ll t,\\ A\,\mathrm{e}^{-A/t^2}, & |k|\gg t.\end{cases}$$

Since $x/(\mathrm{e}^x-1)\leqslant\mathrm{e}^{-x/2}$, $\mathcal{R}_t\leqslant t^2\mathrm{e}^{-A/2t^2}$: at resolution t the noise is no larger than t times a white noise smoothed at scale $1/t$. Pointwise, $\mathrm{Var}\,\eta_t(x)\leqslant\mathcal{R}_t(x,x)\lesssim t^{d+2}$, i.e. $\eta_t(x)=O(t^{1+d/2})$. This is the size at scale $1/t$ of a distribution of regularity $-1-d/2$ ($-5/2$ in $d=3$): seen at its own resolution, the noise is one derivative rougher than white noise ($-d/2$) and two rougher than the GFF ($1-d/2$), the same as the free noise $A\varphi=A^{1/2}\xi$ of the Dyson–Schwinger equation, whatever the interaction. It is not a bound on a limit distribution: η_t is a martingale in t, not the smoothing of a fixed distribution, and on a fixed test function (9.11) only gives $\langle h,\eta_t\rangle=O(t)$, while the free noise stays bounded. For Φ^4_3 no L^2 limit of the logarithmic derivative is expected [ALZ06]. Compared with the free noise, the bound is sharp at and above the resolution, where $\mathcal{R}_t/(A\,\mathrm{e}^{-A/t^2})=(1-\mathrm{e}^{-A/t^2})^{-1}\to 1$, and larger by the factor t^2/A on the resolved modes. Uniformly in the interaction it cannot be improved: for a pinning potential $V_T=\frac{\kappa}{2}\|\varphi\|^2$, the covariance of η_t tends to $\mathcal{R}_t$ as $\kappa\to\infty$ (Exercise 9.5).

Remark 9.6. (Pathwise bounds) (sketch) Work at finite spatial cutoff, with $T=\infty$ or $t\leqslant T/2$, so that $\mathcal{R}_t$ obeys the bounds below uniformly in the cutoff. The norm $\|f\|_\varepsilon=\sup_x(1+|x|)^{-\varepsilon}|f(x)|$ is convex, so Proposition 9.5 dominates $\|\eta_t\|_\varepsilon$ by $\|\Xi\|_\varepsilon$, with $\Xi\sim N(0,\mathcal{R}_t)$. For this Gaussian field, $\mathcal{R}_t(x,x)\lesssim t^{d+2}$ and $\langle\delta_x-\delta_y,\mathcal{R}_t(\delta_x-\delta_y)\rangle\lesssim t^{d+4}|x-y|^2$, so Dudley's bound and Gaussian concentration give Gaussian tails for $\|\Xi\|_\varepsilon$ at the level $t^{1+d/2}\sqrt{\log(2+t)}$, uniformly in L. Since η is a martingale, $\|\eta_s\|_\varepsilon$ is a submartingale, and Doob's inequality extends the tails to $\sup_{s\leqslant t}$. Summing over dyadic t, almost surely

$$\|\eta_t\|_\varepsilon\leqslant C(\omega)\,t^{1+d/2}\sqrt{\log(2+t)}\qquad\text{for all } t\geqslant 1\quad(1\leqslant t\leqslant T/2 \text{ if } T<\infty),$$

with C having Gaussian tails uniformly in the interaction and in the spatial cutoff. This is a pathwise bound at each resolution, in the sense above.

Control of the force. Write $\mathrm{d}F_s = \Upsilon_s \mathrm{d}W_s$ as in Section 8. By Theorem 9.1, $\mathrm{d}\eta_s = (K\dot{Q}_s^{1/2} - J_s\Upsilon_s)\,\mathrm{d}W_s$, so $\int_0^t J_s\Upsilon_s\,\mathrm{d}W_s = K\int_0^t \dot{Q}_s^{1/2}\mathrm{d}W_s - \eta_t$. Itô's isometry, $(a+b)^2 \leqslant 2a^2+2b^2$ and (9.11) give

$$\mathbb{E}\int_0^t \|\Upsilon_s^* J_s h\|^2\,\mathrm{d}s \leqslant 2\,\langle h, \mathcal{R}_t h\rangle + 2\,\langle h, KQ_tKh\rangle. \tag{9.12}$$

For the heat profile with $T=\infty$, $KQ_tK = A\,\mathrm{e}^{-A/t^2} \leqslant t^2/\mathrm{e}$, so the right-hand side is at most $2\,(1+\mathrm{e}^{-1})\,t^2\|h\|_{L^2}^2$. This controls a smoothed derivative of the force, $\Upsilon_s = \mathrm{D}F_s(\Phi_s)\,\dot{Q}_s^{1/2}$, without any estimate on the terminal force.

Consequence for Φ^4_3. In the field equation (9.7) the noise is therefore controlled a priori at each fixed resolution, uniformly in the ultraviolet cutoff. A bound on Φ_t uniform in the ultraviolet cutoff then follows from a coercivity estimate for the renormalised effective force F_t, which converts the control of $A\,\Phi_t - A\,Q_tF_t$ into control of Φ_t. This is the remaining analytic input, and it is where the large-field problem of Lecture 5 enters. Lecture 6 carries this out: its forward estimate is this field equation, with the bound of Proposition 9.5.

9.4 Infinite volume

On $\mathbb{T}_L^d$ the free energy $V_0(0)$ and the effective potential V_t are extensive: they grow like L^d, and (7.9) has no limit as $L\to\infty$. Three objects are intensive and survive formally:

i. *the force.* For a local interaction $V_T(\varphi) = \int v_T(\varphi(x))\mathrm{d}x$, $\mathrm{D}V_T(\varphi)(x) = v_T'(\varphi(x))$, and (7.7) reads, on $\mathbb{R}^d$,

$$\Phi_t = \int_0^t \dot{Q}_s F_s\,\mathrm{d}s + \int_0^t \dot{Q}_s^{1/2}\mathrm{d}W_s, \qquad F_t(x) = -\mathbb{E}_t[v_T'(\Phi_T(x))]. \tag{9.13}$$

Here every quantity is local, and the equation makes sense without a volume cutoff. Infinite-volume limits of $\nu_T^{(L)}$ are expected to be laws of solutions of (9.13). Uniqueness of such solutions (e.g. for small coupling, or by contraction in weighted spaces) is a uniqueness statement for the infinite-volume theory. Non-uniqueness is expected in the presence of phase transitions, e.g. for Φ^4_2 with a deep double-well potential [GJS75];

ii. *the Burgers flow* (8.1) of the force, for the same reason;

iii. *differences of free energies.* For a local f, $\mathcal{U}_T^{(L)}(f)$ in (7.12) is a difference of two extensive infima, and it stays bounded as $L\to\infty$. Its variational characterisation, through the infinite-volume Euler–Lagrange equation (9.13), is the basis of the infinite-volume analysis of Φ^4_2 and $\exp(\Phi)_2$ in [BG23].

Differences of free energies. Fix the volume, and let u^* be the optimal drift of (7.9), so that $X^{u^*} = \Phi$ is the Polchinski process (Theorem 7.5). In (7.12) the infimum may be taken over perturbations $u = u^* + w$ of the optimal drift. The drift enters linearly, so $X^{u^*+w} = \Phi + I(w)$ with $I_t(w) = \int_0^t \dot{Q}_s^{1/2} w_s\,\mathrm{d}s$, and by the Euler–Lagrange equation (7.11) the cross term of the energy is $\mathbb{E}\int_0^T \langle u_s^*, w_s\rangle\,\mathrm{d}s = -\mathbb{E}\langle \mathrm{D}V_T(\Phi_T), I_T(w)\rangle$. The two extensive infima then cancel exactly, and what is left is a variational problem for the perturbation alone.

Proposition 9.7. (Variational principle relative to the FBSDE) *(formal) Let Φ be the Polchinski process, i.e. the solution of the FBSDE (7.7) driven by the Brownian motion W, and ν_T the law of Φ_T. For every f bounded from below,*

$$-\log\int \mathrm{e}^{-f}\,\mathrm{d}\nu_T = \inf_w \mathbb{E}\Big[f(\Phi_T + I_T(w)) + B_{V_T}(\Phi_T, I_T(w)) + \frac{1}{2}\int_0^T \|w_s\|^2\,\mathrm{d}s\Big], \tag{9.14}$$

where the infimum runs over adapted w with $\mathbb{E}\int_0^T \|w_s\|^2 \,\mathrm{d}s < \infty$, and

$$B_V(\phi,h) := V(\phi+h) - V(\phi) - \langle \mathrm{D}V(\phi), h\rangle$$

is the remainder of the first-order Taylor expansion of V. The minimiser satisfies $w_s = -\dot{Q}_s^{1/2}\,\mathbb{E}_s[\mathrm{D}f(\Phi_T^f) + \mathrm{D}V_T(\Phi_T^f) - \mathrm{D}V_T(\Phi_T)]$, with $\Phi^f = \Phi + I(w)$: the perturbed field solves the FBSDE with terminal potential $V_T + f$, and its force is that of Φ corrected by $F_s^f - F_s = -\mathbb{E}_s[\mathrm{D}f(\Phi_T^f) + \mathrm{D}V_T(\Phi_T^f) - \mathrm{D}V_T(\Phi_T)]$.

Proof. By (7.12), the left-hand side is $\inf_u J_f(u) - J_0(u^*)$, with

$$J_f(u) = \mathbb{E}\Big[f(X_T^u) + V_T(X_T^u) + \frac{1}{2}\int_0^T \|u_s\|^2 \,\mathrm{d}s \Big].$$

Writing $u = u^* + w$ and expanding $\frac{1}{2}\|u^*+w\|^2$, the terms $V_T(\Phi_T)$ and $\frac{1}{2}\int\|u^*\|^2$ cancel against $J_0(u^*)$, and the cross term is computed above. The first-order condition in w is obtained as in (7.11). □

For a local interaction $V_T(\varphi) = \int v_T(\varphi(x))\,\mathrm{d}x$,

$$B_{V_T}(\phi,h) = \int b_T(\phi(x),h(x))\,\mathrm{d}x, \qquad b_T(a,h) = v_T(a+h) - v_T(a) - v_T'(a)\,h,$$

and $b_T(a,0) = 0$. So every term of (9.14) is local: there is no free energy and no extensive quantity, only the field Φ and the perturbation that f induces on it. For f supported in a bounded region one expects the optimal w, and with it $I(w)$, to decay away from the support, at the rate of the decay of correlations of the measure. Both sides of (9.14) then make sense in infinite volume, with Φ a solution of the infinite-volume FBSDE (9.13). This suggests the following formal description of the infinite-volume measures (at a fixed ultraviolet cutoff T):

an infinite-volume measure is the law of Φ_T for a solution Φ of (9.13) which satisfies (9.14), with $B = \int b_T\,\mathrm{d}x$, for all local f.

The Laplace functionals in (9.14) determine the measure, and the condition involves only intensive quantities. When the FBSDE has several solutions, as is expected in the presence of phase transitions, each solution gives a measure with its own variational principle; the perturbations w of finite energy only probe a given solution locally. For Φ^4_3 this remainder term contains the divergent counterterms, and, as for (7.12), the change of variables of [BG20] is needed before (9.14) has bounds uniform in T; in two dimensions, compare the infinite-volume analysis of [BG23].

The coercive estimates must then be *local*: one uses weighted norms and needs bounds whose constants do not depend on the volume. This infrared problem is the analogue, for the scale dynamics, of the "coming down from infinity" estimates for the parabolic dynamics.

Exercises

Exercise 9.1. (Doob transform) Show that G_t in (9.1) solves (9.2), by writing $\mathrm{e}^{-V_t}G_t = \mathrm{e}^{\frac{1}{2}\Delta_{Q_{T,t}}}(\mathrm{e}^{-V_T}G)$. Prove (9.3).

Exercise 9.2. (The relative variational principle) (a) Derive (9.14) and the equation for the minimiser. (b) For a Gaussian ν_T (quadratic V_T) and a linear $f = \langle g, \cdot\rangle$, compute the remainder term, solve (9.14), and recover $-\log\int \mathrm{e}^{-\langle g,\cdot\rangle}\,\mathrm{d}\nu_T = \langle g, m\rangle - \frac{1}{2}\langle g, \Sigma g\rangle$, with m and Σ the mean and covariance. (c) For a local interaction and f supported in a ball, show that B_{V_T} only sees the region where $I(w) \neq 0$, and discuss why (9.14) is bounded uniformly in the volume.

Exercise 9.3. (DS from the dynamics) Prove (9.5) without using Proposition 7.2, for G of the form $G_t(\Phi_t)$ with G_t solving (9.2). Show that the drift of $D_{J,h}G_t(\Phi_t)$ equals the covariation of $G_t(\Phi_t)$ with $\langle h, \eta_t\rangle$. Where is the symmetry of $\mathrm{D}F_t$ used?

Exercise 9.4. (Gaussian check) For $V_T = 0$ show that (9.5) reduces to $\mathbb{E}\,[D_{Q,Kh} G(X_t)] = \mathbb{E}[G(X_t)\langle Kh, X_t\rangle]$, and compare with the DS characterisation of the GFF in Lecture 2.

Exercise 9.5. (Sub-Gaussian noise) (a) Check (9.9) and the bridge representation, and show that the Girsanov density depends only on X_T. (b) Compute $\mathcal{R}_t$ for the heat profile, and show that $\mathcal{R}_t \leqslant \frac{4}{3} t^2$ for $T \geqslant 2t$ and $\mathcal{R}_t \leqslant t^2$ for $T = \infty$. (c) Compute the law of η_t for $V_T = 0$ and for $V_T = \frac{\kappa}{2}\|\varphi\|^2$, and show that its covariance tends to $\mathcal{R}_t$ as $\kappa \to \infty$. In what sense is (9.11) sharp? (d) Show that $\mathcal{R}_t(x,x) \asymp t^{d+2}$ for $t \gg m$, and complete the argument of Remark 9.6.

Appendices: additional material

Appendix A Algebraic structure of the Polchinski flow

A.1 A semigroup of renormalisation maps

For a positive covariance S define the renormalisation map

$$\mathcal{R}_S(V) := -\log\,(\mu^S * \mathrm{e}^{-V}), \qquad (\mu^S * u)(\phi) = \int u\,(\phi+\zeta)\,\mu^S(\mathrm{d}\zeta). \tag{A.1}$$

Since $\mu^S * \mu^{S'} = \mu^{S+S'}$,

$$\mathcal{R}_S \circ \mathcal{R}_{S'} = \mathcal{R}_{S+S'} = \mathcal{R}_{S'} \circ \mathcal{R}_S, \qquad V_t = \mathcal{R}_{Q_{T,t}}(V_T). \tag{A.2}$$

So the effective potentials are an orbit of an *abelian* semigroup, indexed by the cone of positive covariances, acting nonlinearly on interactions and linearly on Boltzmann factors e^{-V}. Its infinitesimal generators are the vector fields

$$\mathcal{D}_B(V) := \tfrac{1}{2}\mathrm{Tr}(B\,\mathrm{D}^2 V) - \tfrac{1}{2}\langle \mathrm{D}V, B\,\mathrm{D}V\rangle, \qquad \tfrac{\mathrm{d}}{\mathrm{d}s}\mathcal{R}_{sB}(V)\Big|_{s=0} = \mathcal{D}_B(V), \tag{A.3}$$

and (7.8) reads $\partial_t V_t = -\mathcal{D}_{\dot{Q}_t}(V_t)$.

Proposition A.1. (Flatness) (formal) *(a)* $[\mathcal{D}_B, \mathcal{D}_{B'}] = 0$ *for all positive* B, B'. *(b)* V_t *depends on the profile only through* $Q_{T,t}$. *In particular the free energy* $V_0(0)$ *and the law* ν_T *depend only on* Q_T, *not on the path* $t \mapsto Q_t$.

Proof. (a) Differentiate $\mathcal{R}_{sB}\,\mathcal{R}_{rB'} = \mathcal{R}_{rB'}\,\mathcal{R}_{sB}$ at $s = r = 0$, or compute directly (Exercise A.1). (b) is (A.2). □

Different profiles with the same endpoints are different *coordinate systems* on the same renormalisation group trajectory. They change the process Φ and the observables at intermediate resolution, but not the terminal theory. The flow has no holonomy [BCG26].

A.2 Perturbation theory and counterterms

Expanding $V_t = \mathcal{R}_{Q_{T,t}}(V_T)$ in cumulants of the Gaussian increment $\zeta \sim \mu^{Q_{T,t}}$ gives

$$V_t(\phi) = \mathbb{E}_\zeta\,[V_T\,(\phi+\zeta)] - \tfrac{1}{2}\mathrm{Var}_\zeta(V_T\,(\phi+\zeta)) + \dots \tag{A.4}$$

For (7.1), Exercise 6.1 of Lecture 3 shows that Gaussian convolution simply *lowers the Wick ordering*: since $Q_{T,t}(x,x) = c_T - c_t$,

$$V_t(\phi) = \int [\lambda H_4(\phi(x); c_t) + \gamma_T H_2(\phi(x); c_t)]\mathrm{d}x + O(\lambda^2). \tag{A.5}$$

Here $\gamma_T = O(\lambda^2)$ is kept for later use. At first order the effective potential is the Wick-ordered potential at the current resolution. This is the dual statement of Proposition 6.1 of Lecture 3: $\int H_4(X_t; c_t)$ is an observable.

Proposition A.2. (Second order) (formal) *Write $S = Q_{T,t}$ and $:\phi^p:_t = H_p(\phi; c_t)$. Then*

$$V_t(\phi) = \int [\lambda :\phi^4:_t + \gamma_T :\phi^2:_t] - \frac{\lambda^2}{2} \sum_{k=1}^{4} \binom{4}{k}^2 k! \iint S(x,y)^k :\phi^{4-k}:_t(x) :\phi^{4-k}:_t(y)\, \mathrm{d}x\, \mathrm{d}y, \tag{A.6}$$

up to $O(\lambda^3)$.

Proof. By the binomial formula for Hermite polynomials,

$$H_4(\phi + \zeta; c_t + s) = \sum_{k=0}^{4} \binom{4}{k} H_k(\zeta; s)\, H_{4-k}(\phi; c_t), \qquad s = S(x,x).$$

Then compute the variance in (A.4) with Lemma 3.8 of Lecture 2. □

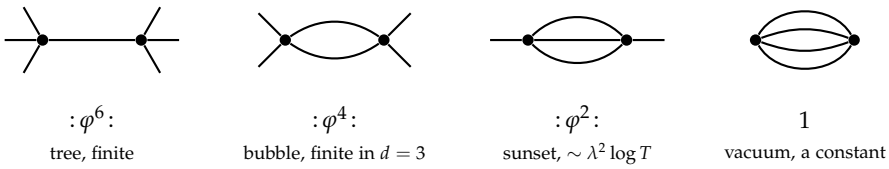


Figure A.1. The terms $k = 1,2,3,4$ of (A.6), with k lines S joining two Wick-ordered vertices; Wick ordering has removed the lines from a vertex to itself. As $T \to \infty$ in $d = 3$, only the sunset, $k = 3$, is a non-integrable divergence for the $:\varphi^2:$ term; it is cancelled by γ_T.

Now let $T \to \infty$ at fixed t, so that $S \to C - Q_t$, which near the diagonal is as singular as G. In $d = 2$ every S^k is locally integrable, so with $\gamma_T = 0$ every term has a limit: Wick ordering suffices. In $d = 3$, $G(z) \sim (4\pi|z|)^{-1}$:

- $k = 1,2$: S and S^2 are locally integrable. These terms are finite and nonlocal.
- $k = 3$ (the *sunset*): $S^3 \sim |z|^{-3}$ is not integrable. Writing $\phi(x)\,\phi(y) = \frac{1}{2}(\phi(x)^2 + \phi(y)^2) - \frac{1}{2}(\phi(x) - \phi(y))^2$, up to a field-independent constant the divergent part is local:

  $$-48\lambda^2 \Big(\int S^3 \Big) \int :\phi^2:_t + \text{(finite)},$$

  with $\int S^3 \sim \frac{1}{16\pi^2} \log T$. It is cancelled by

  $$\gamma_T := 48\lambda^2 \int_{\mathbb{T}_L^3} Q_T(z)^3\, \mathrm{d}z = \frac{3\lambda^2}{\pi^2} \log T + O(1), \tag{A.7}$$

  since $\gamma_T - 48\lambda^2 \int S^3$ converges (Exercise A.2).
- $k = 4$: a divergent constant $-12\lambda^2 L^3 \int S^4 \sim T$, the vacuum energy. It cancels in ν_T but not in the free energy.

Power counting (Exercise A.3) shows that no further field-dependent counterterms are needed in $d = 3$: Φ_3^4 is superrenormalisable. Thus renormalisation appears as a property of the flow: one chooses the terminal data V_T so that the effective potentials V_t at *fixed* resolution have limits. The divergences at second order are also related to the singularity of Φ_3^4 with respect to the GFF [BG21]: the measures ν_t at fixed t are expected to stay absolutely continuous with respect to μ^{Q_t}, while the terminal one is not.

Exercises

Exercise A.1. (Commuting flows) Show by direct computation that $\mathrm{D}\mathcal{P}_B(V)[\mathcal{P}_{B'}(V)] = \mathrm{D}\mathcal{P}_{B'}(V)[\mathcal{P}_B(V)]$. Which property of D^2V and D^3V is used?

Exercise A.2. (The sunset) (a) Show that $\int_{|z|\leqslant 1} G(z)^3\,\mathrm{d}z = \infty$ in $d=3$, and that $\int Q_T^3 = \frac{1}{16\pi^2}\log T + O(1)$ for the heat profile. (b) Show that $Q_T^3 - (Q_T - Q_t)^3$ is integrable uniformly in T, and deduce that the counterterm (A.7) makes the $:\phi^2:$ coefficient in (A.6) converge. (c) Show that the remaining part $\iint S^3(x,y)\,(\phi(x)-\phi(y))^2$ converges for smooth ϕ.

Exercise A.3. (Power counting) Show that the superficial degree of divergence of a connected graph of Φ_d^4 with n vertices and E external legs is $\omega = d - \frac{d-2}{2}E + (d-4)\,n$. Deduce that in $d=3$ the only divergent graphs with loops are the two-point graphs with $n\leqslant 2$ (the tadpole and double tadpole, removed by Wick ordering, and the sunset) and vacuum graphs with $n\leqslant 3$.

Appendix B Stochastic localisation

Proposition 7.2 and Section 7.3 say that the Polchinski process is a stochastic localisation of ν_T. This appendix records why that decomposition is useful, and what it proves.

Why it is useful. Three features make this decomposition a tool:

- *It is a martingale.* $\nu_T = \mathbb{E}\,\mu_t^{\Phi_t}$, and the expectation of any test function under $\mu_t^{\Phi_t}$ is a martingale. Itô calculus then splits variance and entropy exactly into the part already revealed and the part still hidden.
- *The pieces are simpler.* The Gaussian factor adds $\mathcal{R}_t$ to the Hessian of the potential. If ν_T is log-concave, μ_t^{ϕ} is $\mathcal{R}_t$-uniformly log-concave, so Bakry–Émery and Brascamp–Lieb apply to it even when their constants are useless for ν_T itself.
- *One quantity controls the loss.* What is revealed per unit resolution is governed by Cov_t, whose evolution (8.9) is explicit. An inequality for ν_T reduces to showing that the localisation does not happen too fast.

The process can also be run as an algorithm: with an approximate force it is a sampler, as in the algorithmic stochastic localisation of [EMS22], and denoising diffusions are of the same kind [Mon23]. By (9.9), the noise of the field equation is, up to J_t, the score that such algorithms estimate.

B.1 Log-Sobolev inequalities along the localisation

Bauerschmidt and Bodineau [BB21] proved log-Sobolev inequalities by running the Polchinski flow (see the survey [BBD24]). In their language the renormalised measure at a given scale is the law of Φ_t, and the fluctuation measure is the conditional law $\mu_t^{\Phi_t}$ of Proposition 7.2, i.e. the stochastic localisation of ν_T [Eld23]. (Their scale parameter runs opposite to our resolution: their initial scale is our T, where the renormalised measure is ν_T and the fluctuation measure is trivial, and their integrated covariance is our S_t.) In this dictionary their argument is an entropy computation along our filtration.

Entropy as quadratic variation. Let $\rho \geqslant 0$ with $\mathbb{E}_{\nu_T}\rho = 1$, and let $M_t = \mathbb{E}_t[\rho(\Phi_T)] = G_t(\Phi_t)$, with $G_t(\phi) = \int \rho\,\mathrm{d}\mu_t^{\phi}$. The martingale M runs from $M_0 = 1$ to $M_T = \rho(\Phi_T)$, and its noise is that of the field: $\mathrm{d}M_t = \langle \mathrm{D}G_t(\Phi_t), \dot{Q}_t^{1/2}\,\mathrm{d}W_t\rangle$. Itô's formula for $M\log M$ gives

$$\mathrm{Ent}_{\nu_T}(\rho) = \frac{1}{2}\,\mathbb{E}\int_0^T \frac{\mathrm{d}\langle M\rangle_t}{M_t} = 2\,\mathbb{E}\int_0^T \big\|\dot{Q}_t^{1/2} D\sqrt{G_t}(\Phi_t)\big\|^2\,\mathrm{d}t,$$

and, stopping at resolution t, the chain rule

$$\mathrm{Ent}_{\nu_T}(\rho) = \mathrm{Ent}(G_t(\Phi_t)) + \mathbb{E}\big[\mathrm{Ent}_{\mu_t^{\Phi_t}}(\rho)\big]:$$

the entropy already revealed at resolution t plus the expected entropy still hidden. The increments are controlled by the conditional law: differentiating the density of μ_t^{ϕ} in ϕ, $\mathrm{D}G_t(\phi) = S_t^{-1}\mathrm{Cov}_{\mu_t^{\phi}}(\rho, y)$.

Theorem B.1. (Multiscale Bakry–Émery [BB21, BBD24]) *(formal in infinite dimensions) Suppose that there are real numbers (possibly negative) κ_t such that, for all $t\in[0,T]$ and all ϕ, as quadratic forms,*

$$\dot{Q}_t\,\mathrm{D}^2V_t(\phi)\,\dot{Q}_t + \frac{1}{2}\ddot{Q}_t \geqslant \kappa_t\dot{Q}_t.$$

Then ν_T satisfies the log-Sobolev inequality

$$\mathrm{Ent}_{\nu_T}(\rho) \leqslant \frac{2}{\gamma}\,\mathbb{E}_{\nu_T}[\|\dot{Q}_T^{1/2} D\sqrt{\rho}\|^2], \qquad \frac{1}{\gamma} = \int_0^T \exp\Big(-2\int_t^T \kappa_r\,\mathrm{d}r\Big)\,\mathrm{d}t.$$

The hypothesis gives, at each resolution, the gradient bound

$$\|\dot{Q}_t^{1/2} D\sqrt{G_t}\|^2 \leqslant \mathrm{e}^{-2\int_t^T \kappa_r \mathrm{d}r} \int \|\dot{Q}_T^{1/2} D\sqrt{\rho}\|^2\,\mathrm{d}\mu_t^{\phi};$$

inserted in the entropy formula it gives the theorem [BBD24]. For $V_T = 0$ and a linear profile $Q_t = t\,Q_T/T$ one may take $\kappa = 0$, so $1/\gamma = T$ and the theorem is the Gaussian log-Sobolev inequality $\mathrm{Ent}(\rho) \leqslant 2\,\mathbb{E}\langle D\sqrt{\rho}, Q_T D\sqrt{\rho}\rangle$. The hypothesis is a convexity condition on the effective potentials at every scale, not on V_T, corrected by the scale dependence $\ddot{Q}_t$ of the decomposition; κ_t may be negative at some scales: only the weights $\mathrm{e}^{-2\int_t^T \kappa_r \mathrm{d}r}$ enter γ.

In terms of the conditional covariance. By Proposition 8.3, $\mathrm{D}^2V_t = S_t^{-1} - S_t^{-1}\mathrm{Cov}_t\,S_t^{-1}$, and $S_t^{-1}\dot{Q}_t S_t^{-1} = \dot{\mathcal{R}}_t$ is the derivative of the regulator (8.6). Since $\dot{Q}_t S_t^{-1} = S_t\,\dot{\mathcal{R}}_t$, the hypothesis reads

$$S_t\,\dot{\mathcal{R}}_t\mathrm{Cov}_t\,\dot{\mathcal{R}}_t S_t \leqslant \dot{Q}_t S_t^{-1}\dot{Q}_t + \frac{1}{2}\ddot{Q}_t - \kappa_t\dot{Q}_t.$$

If $\dot{Q}_t$ is invertible, multiplying on the left by $S_t\dot{Q}_t^{-1} = \dot{\mathcal{R}}_t^{-1}S_t^{-1}$ and on the right by its transpose $\dot{Q}_t^{-1}S_t$ shows that it is equivalent to the explicit bound

$$\mathrm{Cov}_{\mu_t^{\phi}} \leqslant S_t - S_t\dot{Q}_t^{-1}\Big(\kappa_t\dot{Q}_t - \frac{1}{2}\ddot{Q}_t\Big)\dot{Q}_t^{-1}S_t \qquad \forall\phi. \tag{B.1}$$

The second derivative $\ddot{Q}_t$ is not estimated: it is a property of the Gaussian decomposition, and it enters only through the combination $\kappa_t\dot{Q}_t - \frac{1}{2}\ddot{Q}_t$. When Q_t and S_t are functions of one operator (Fourier multipliers, say), the right-hand side is $S_t - S_t^2\dot{Q}_t^{-2}(\kappa_t\dot{Q}_t - \frac{1}{2}\ddot{Q}_t)$ mode by mode. If $\kappa_t\dot{Q}_t > \frac{1}{2}\ddot{Q}_t$ (for the linear profile: $\kappa_t > 0$), every conditional law is strictly more concentrated than the unresolved Gaussian fluctuation; if not, Cov_t may exceed S_t, as in a double well at low resolution. A concave profile ($\ddot{Q} \leqslant 0$) makes the bound smaller but forces κ down by the same amount: for the Gaussian with $Q_t = 1 - \mathrm{e}^{-t}$ one has $\ddot{Q} = -\dot{Q}$, the hypothesis holds with $\kappa = -\frac{1}{2}$, and $\mathrm{Cov}_t = S_t$. The hypothesis is thus an upper bound on the conditional covariance, uniform in the conditioning, measured in the metric $\dot{\mathcal{R}}_t$. Closely related is the *entropic stability* of Chen and Eldan [CE22] (see also [BBD24]): if, for all $r \leqslant t$ and all ϕ, $\dot{\mathcal{R}}_r\mathrm{Cov}_{\mu_r^{\phi}}\dot{\mathcal{R}}_r \leqslant \alpha_r\dot{\mathcal{R}}_r$, then

$$\mathrm{Ent}_{\nu_T}(\rho) \leqslant \exp\Big(\int_0^t \alpha_r\,\mathrm{d}r\Big)\,\mathbb{E}\big[\mathrm{Ent}_{\mu_t^{\Phi_t}}(\rho)\big]:$$

the total entropy is controlled by the entropy that is still hidden at resolution t. The low-resolution part of the localisation is where the measure may be multimodal, and the condition says that it does not hide too much. With the Brascamp–Lieb bounds of Section 8.4 the conditional covariance is controlled by the Hessian of V_T when this is convex; for Φ_3^4 it is not, and the condition is a statement about the effective potentials, i.e. about the renormalisation group flow itself.

Proposition B.2. (Poincaré from the localisation) *Suppose that, for all $r \leqslant t$ and all ϕ, $\dot{\mathcal{R}}_r \mathrm{Cov}_{\mu_r^\phi} \dot{\mathcal{R}}_r \leqslant \alpha_r \dot{\mathcal{R}}_r$. Then, for every f,*

$$\mathrm{Var}_{\nu_T}(f) \leqslant \exp\Big(\int_0^t \alpha_r \,\mathrm{d}r\Big)\, \mathbb{E}\,\mathrm{Var}_{\mu_t^{\Phi_t}}(f).$$

If moreover ν_T is log-concave, then

$$\mathrm{Var}_{\nu_T}(f) \leqslant \exp\Big(\int_0^t \alpha_r \,\mathrm{d}r\Big)\, \mathbb{E}_{\nu_T}\langle \mathrm{D}f, \mathcal{R}_t^{-1}\mathrm{D}f\rangle.$$

Proof. Let $M_r = \mathbb{E}_r f(\Phi_T)$ and $\chi_r = \mathrm{Cov}_{\mu_r^{\Phi_r}}(f, y)$. As in Section B.1, $\mathrm{d}\langle M\rangle_r = \langle \chi_r, \dot{\mathcal{R}}_r \chi_r\rangle \,\mathrm{d}r$, so $v(r) := \mathbb{E}\,\mathrm{Var}_{\mu_r^{\Phi_r}}(f)$ has $v' = -\mathbb{E}\langle \chi_r, \dot{\mathcal{R}}_r \chi_r\rangle$ and $v(0) = \mathrm{Var}_{\nu_T}(f)$. By Cauchy–Schwarz, with $h = \dot{\mathcal{R}}_r \chi_r$, $\langle \chi_r, h\rangle^2 \leqslant \mathrm{Var}_{\mu_r^{\Phi_r}}(f)\,\langle h, \mathrm{Cov}_r h\rangle \leqslant \alpha_r \,\mathrm{Var}_{\mu_r^{\Phi_r}}(f)\,\langle \chi_r, \dot{\mathcal{R}}_r \chi_r\rangle$. Hence $v' \geqslant -\alpha_r v$, which gives the first bound. For the second, the Hessian of the potential of μ_t^ϕ is at least $\mathcal{R}_t$: apply Brascamp–Lieb to μ_t^ϕ and take the expectation. □

For $V_T = 0$ one has $\mathrm{Cov}_r = S_r$. In one mode, or for the linear profile, where $\dot{Q}_r S_r^{-1}$ is a number, $\alpha_r = \dot{Q}_r S_r^{-1}$ and $\exp\int_0^t \alpha_r \,\mathrm{d}r = Q_T S_t^{-1}$: the first bound is an equality for linear f, and the second tends to the Gaussian Poincaré inequality as $t \to T$ (Exercise B.2).

An application: the KLS conjecture. Take an isotropic log-concave measure on $\mathbb{R}^n$ (mean zero, covariance 1) and the identity control, $\mathcal{R}_t = t$. The second bound reads $\mathrm{Var}(f) \leqslant t^{-1}\,\mathrm{e}^{\int_0^t \alpha}\,\mathbb{E}|\mathrm{D}f|^2$, so a Poincaré inequality follows if Cov_r stays bounded in operator norm up to some time t. The Kannan–Lovász–Simonovits conjecture asks for a Poincaré constant independent of n, and hence for a time t independent of n. Along the path, the bound holds only with high probability, not uniformly in ϕ, so the argument has to be adapted, and proving that the covariance stays bounded is the heart of the matter. Following Eldan [Eld13], this has given the bounds of Chen [Che21] and Klartag–Lehec [KL22], and then Klartag's bound [Kla23]: the KLS isoperimetric constant is at most $C\sqrt{\log n}$, i.e. the Poincaré constant is $O(\log n)$. The related slicing conjecture has since been proved [KL24].

The connection with Bauerschmidt–Bodineau. The multiscale Bakry–Émery criterion of Theorem B.1 and the localisation study the same measure-valued martingale from two sides. Bauerschmidt and Bodineau [BB21] (see [BB19] for an earlier decomposition argument, and [BBD24] for a survey) start from the renormalisation group: integrate out the fine scales and follow the potential. Eldan's approach follows, along one path, the conditional law. The dictionary is:

this lecture	Bauerschmidt–Bodineau	stochastic localisation
resolution t increasing	their scale, reversed	time t increasing
law ν_t of Φ_t	renormalised measure	law of the tilt
μ_t^ϕ	fluctuation measure	tilted measure
S_t, $\mathcal{R}_t$	integrated covariance, —	—, t
$G_t = \int \rho \,\mathrm{d}\mu_t^\phi$	Polchinski semigroup on ρ	martingale $\int \rho \,\mathrm{d}\mu_t$
$\mathrm{D}^2 V_t = S_t^{-1} - S_t^{-1}\mathrm{Cov}_t S_t^{-1}$	Hessian of the potential	covariance of the tilted measure

Proposition 8.3 is the bridge: convexity of the renormalised potentials is an upper bound on the conditional covariance, namely (B.1). The two methods then use this information differently.

- *Bauerschmidt–Bodineau.* The Kolmogorov equation (9.2) transports G_t by the drift $\dot{Q}_t F_t = -\dot{Q}_t \mathrm{D} V_t$. Its Jacobian plays the role of the curvature in the Bakry–Émery computation, and the scale dependence of the metric $\dot{Q}_t$ adds $\frac{1}{2}\ddot{Q}_t$. This yields the gradient bound for G_t at each scale, which is integrated in the entropy formula. The output is a log-Sobolev inequality with an explicit constant, directly at the terminal scale, also far from log-concavity. The hypothesis is uniform in ϕ, which suits field theories where the renormalisation group controls V_t for all fields, as for sine-Gordon at $\beta < 6\pi$ [BB21].
- *Localisation.* One bounds what is lost on $[0,t]$ (the proposition above, or entropic stability [CE22]) and uses a known inequality for μ_t^ϕ. It works directly with moments of the conditional law, and can use bounds that hold only with high probability along the path, as in the KLS problem.

Exercises

Exercise B.1. (**Entropy along the localisation**) (a) Derive the formula for $\mathrm{Ent}_{\nu_T}(\rho)$ as the integrated quadratic variation of M, and the chain rule. (b) Show that $\mathrm{D} G_t(\phi) = S_t^{-1}\mathrm{Cov}_{\mu_t^\phi}(\rho, y)$. (c) For $V_T = 0$ and a linear profile, recover the Gaussian log-Sobolev inequality from Theorem B.1. (d) In zero dimensions, with $Q_t = 1 - \mathrm{e}^{-t}$ and a convex V_T, find κ_t and the resulting constant, and compare with the Bakry–Émery criterion.

Exercise B.2. (**Poincaré from the localisation**) (a) Check that $N_t = S_t^{-1}\zeta_t$ has covariance $\mathcal{R}_{s\wedge t}$. (b) For $V_T = 0$ and one mode, compute α_r and verify the Gaussian check above. (c) In zero dimensions, with $\mathcal{R}_r = r$ and $\nu = \frac{1}{2}N(a,\sigma^2) + \frac{1}{2}N(-a,\sigma^2)$, compute $\mathrm{Cov}_{\mu_r^\phi}$ and its supremum over ϕ. Using the first bound as $t \to \infty$, deduce the Poincaré constant $\leqslant \sigma^2 \mathrm{e}^{a^2/\sigma^2}$. Compare with the true order $\mathrm{e}^{a^2/2\sigma^2}$: why is the exponent off by a factor 2?

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